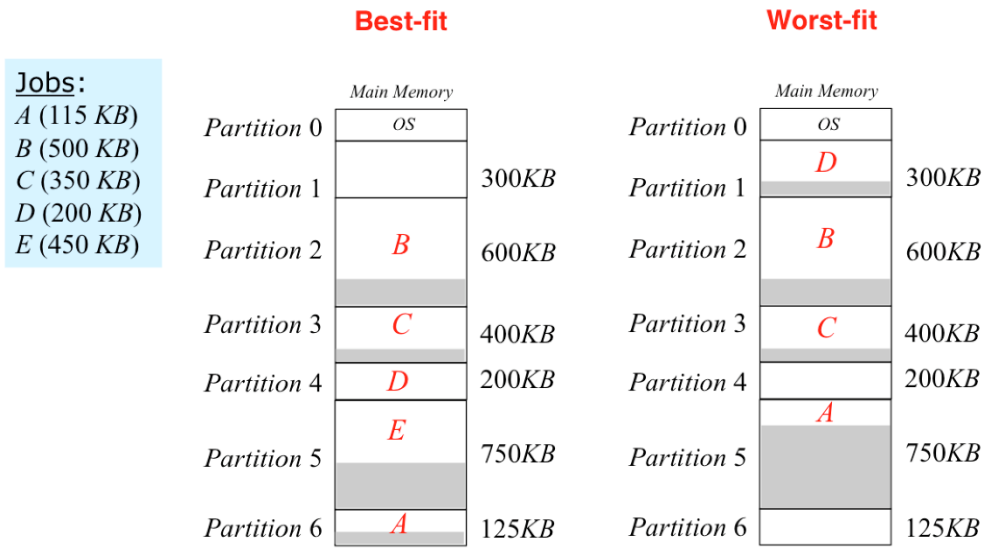
INSTRUCTION: Please answer all questions in the answer booklet. Show all your workings.

1. **[20 MARKS]** There are 5 processes (in order) of *A* (115 KB), *B* (500 KB), *C* (350 KB), *D* (200 KB), and *E* (450 KB) need to be placed in the memory.

   a) Suppose a fixed-partition is used with six memory partitions of 300 KB, 600 KB, 400 KB, 200 KB, 750 KB, and 125 KB (in order).

      (i) How would the best-fit and worst-fit algorithms place the processes? Draw the memory stack and label the partition size clearly. [10m]

Answer : Fixed-partition: Exercises 8.3 (page: 384)



      (ii) In terms of process placement in the memory, which algorithm faced a problem? What is the problem and state the cause of the problem. [4m]

Worst-fit.

Problem: Process E cannot be placed in the memory for Worst-fit

Cause: since there is no enough available partition.

   b) **[10 MARKS]** Suppose a variable or dynamic partition is used with the same 5 processes in Question 1. The following events happened:

t_1: process A and D left.

t_2: process F (50K) enters

t_3: process B left.

t_4: process G (550K) enters.

Draw the memory stack at t_2, t_3 and t_4 and label the sizes clearly for all processes. Use first-fit algorithm and relocatable dynamic partitions of merging adjacent holes to place the process in the memory. [6m]

F	50
	65
B	500
C	350
	200
E	450

T2

F	50
	565
C	350
	200
E	450

T3

F	50
G	550
	15
C	350
	200
E	450

T4

2. **[10 MARKS]** Answer the following questions.

a) Consider a logical address space of 2^8 pages with a 4 *KB* page size, mapped onto a physical memory of 4096 frames. Its page table is given in Table 2.

Table 2

0	50
1	30
2	400
3	100

(i) What is the size of logical and physical addresses (in 2^n bytes)? [4m]

page size → 4 KB = 2^{12}

- [2m] Logical address = totalpage x page size = 2^8 x 2^{12} = 2^{20} bytes
- [2m] Physical address = totalframe x frame size = (4096 pages) x (frame size = page size) =2^{12} x 2^{12} = 2^{24} bytes

(ii) Calculate the physical address if the logical address is 3 : 400. [3m]

3 → 100. Frame size = page size

Physical address = (frame# x frame size) + offset

= (100 x 2^{12}) + 400

= (100 x 4096) + 400

= 4096000 + 400

= 4096400

(iii) What is the page number and offset for the logical address of 51152? [3m]

51152/4096 = 12.48828125

Page Number: 12 – 1 = 11

Offset = (12.48828125 – 12) x 4096 = 2000 – 1 = 1999

3. **[12 MARKS]** Define the following terms.

   a) Demand paging [3m]

   Load pages only needed in virtual memory

   b) Lazy swapper

   Used in demand paging. Swaps a page into memory only when needed.

   c) Belady's anamoly [3m]

   Page fault rate may increase as the number of allocated frames increases, even though it looks like it should be the opposite effect.

   d) Thrashing in virtual memory [3m]

   High paging activity. Spending more time paging than executing.

4. **[13 MARKS]** Consider the following page reference string: **7, 2, 3, 1, 2, 5, 3, 4, 6, 7, 1, 3.** Assuming demand paging with four frames, how many page faults would occur for the following replacement algorithms? Please circle which references cause page faults.

   a) Optimal (OPT) replacement [6.5m]

7	2	3	1	2	5	3	4	6	7	1	3
7	7	7	7		7		7	7			
	2	2	2		5		4	6			
		3	3		3		3	3			
			1		1		1	1			

7 page fault

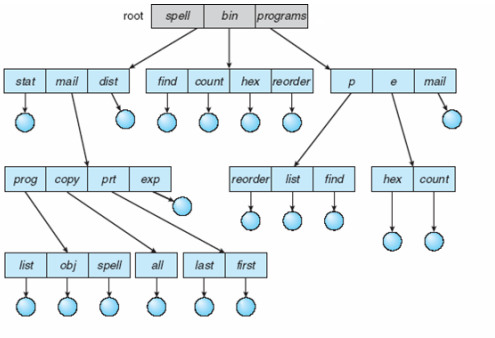
   b) LRU replacement [6.5m]

7	2	3	1	2	5	3	4	6	7	1	3
7	7	7	7		5		5	5	7	7	7
	2	2	2		2		2	6	6	6	6
		3	3		3		3	3	3	1	1
			1		1		4	4	4	4	3

10 page fault

5. **[10 MARKS]** Draw and discuss the advantages of:

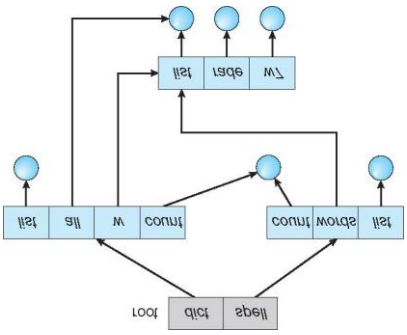
   a) Tree-structured directories [5m]



Text book 10.3.5, slide 13.21- 13.24

- Extention of two-level directory, tree can have arbitrary height
- Separate directory for each user – they can create their own subdirectory and organize their file accordingly.
- The tree has a root directory and every file has unique path name,
- Can have the same file name for different user
- Efficient searching
- Grouping capability

b) Acyclic-graph directories [5m]



Text book 10.3.6, slide 13.25

- Files/subdirectories have two different names (aliasing)
    - Only one actual file exists, so any changes made by one person are immediately visible to the other.
- How do we implement shared files and subdirectories?
    - Can duplicate the *inode* information about the shared file/subdirectory in each of the subdirectories that "point' to the shared structure.
        - If some information changes about the file it had to be updated in several places.
    - Have a new directory entry type:
        - **Link** – another name (pointer) to an existing file or subdirectory.

6. [**10 MARKS]** Answer the following questions:

a) In the Unix system, directory listing is shown as below:

```
-rw-r--r--      1 pbg staff      9423    Apr 19 2018   program
drwx------      3 pbg staff      1024    Aug 19 10:31  mail/
```

(i) Name and define the directory protection. [3m]

Directory name mail/
owner only allows to read write execute

(ii) Name and define the file protection. [3m]

Program owner can read and write ,
group can read only,
universe can read only

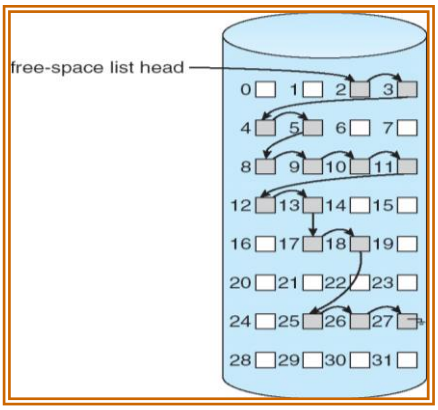
(iii) How to let all the users use the file without restriction? [3m]

chmod 777 program

7. **[10 MARKS]** Describe and show the example of free space management using:

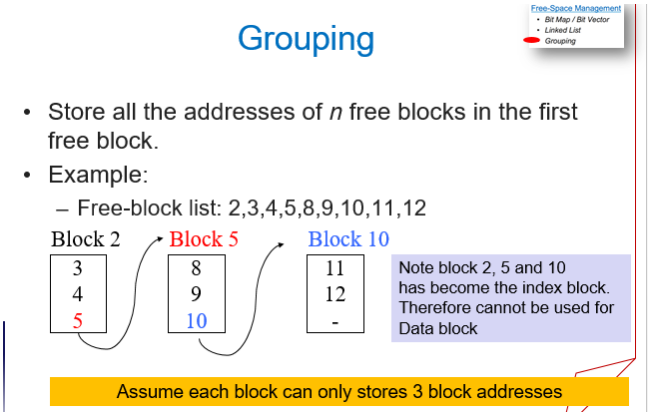
a) Linked list [5m]

To link together all the free disk block. Keeping a pointer to the first free block in a special location on the disk and caching it in memory.



b) Grouping [5m]

The first n-1 of these blocks are actually free. The last block contains the addresses of another n free blocks and so an. The addresses of a large number of free blocks can now be found quickly, unlike the linked-list

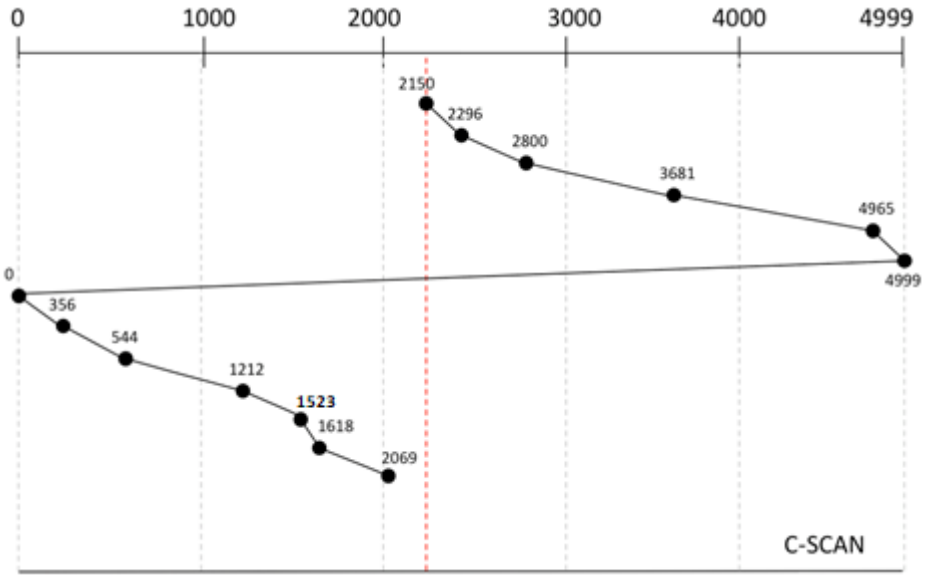


8. **[15 MARKS]** Consider the following information:

- A disk drive has 5000 cylinders, numbered 0 – 4999.
- Using C-SCAN disk-scheduling algorithm to satisfy all the cylinder requests.
- The drive is currently serving a request at cylinder 2150.
- The queue of pending requests is: **2069, 1212, 2296, 2800, 544, 1618, 356, 1523, 4965, 3681**

Starting from the current head position, answer the following question.

a) Draw the whole disk arm movement to satisfy all the pending requests. [5m]



b) What is the total seek distance (in cylinder) that the disk arm had moved? [2m]

4999 - 2150 = 2849

4999 - 0 = 4999

2069 - 0 = 2069

Total distance = 2849 + 4999 + 2069 = 9917 cylinders

c) Now, if C-LOOK algorithm is used to satisfy all the cylinder pending requests, compare the speed of both disk-scheduling algorithms based on the distance (in cylinder) that the disk arm had moved. [3m]

- C-LOOK uses total distance 780 less than S-SCAN
  - (4999-4965)x2 = 34 x 2 = 68
  - (356 x 2) = 712
  - total less = 68 + 712 = 780
- therefore, C-LOOK faster than S-SCAN

Time left 0:43:58

Question 6

Not yet answered

Marked out of 12.00

1. Figure 1 below illustrates the operation of channel I/O. Suppose the system has a request to transfer data from Tape X to Printer Y, analyze the figure below and formulate the path of communication of the transfer operation. Assume that CPU has placed the I/O program in the memory and the IOP is ready to fetch the instruction.

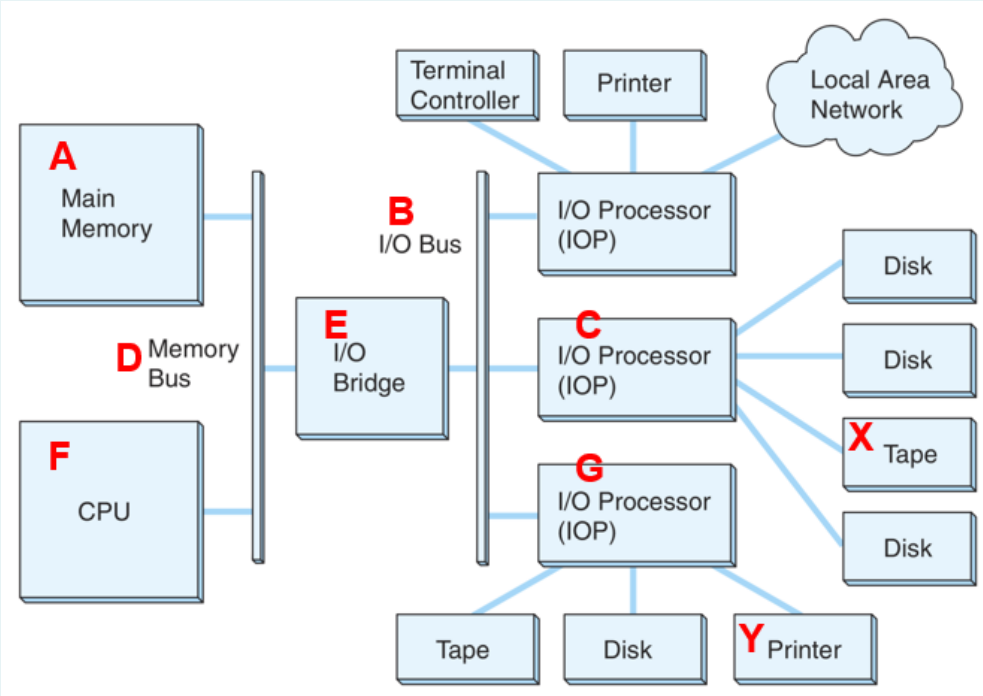


Figure 1: Channel I/O

Answer format: Write the path of communication using only the label in red separated by a comma with no white space in between, e.g. Printer Y communicating with I/O bus through IOP is written as Y,G,B

IOP fetches instruction for transfer operation of Tape X

IOP execute the transfer to Printer Y

2. Consider a server with RAID Level 3 and given Figure 2 below, determine the bit for data regeneration.

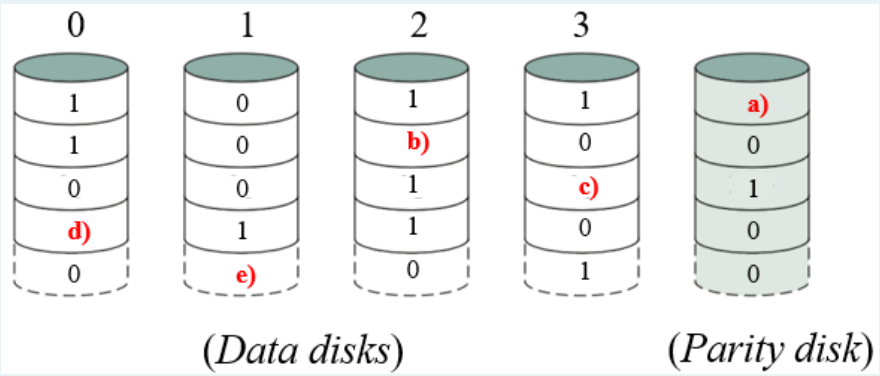


Figure 2: Server with RAID 3

a)

b)

c)

d)

e)

3. Suppose there are 6 data drives and they need to be stored as RAID 5, calculate how many drives are needed altogether.

The drives required is determined by with N is the . Therefore,

drives are needed.

4. Suppose a system has 5 drives as in Figure 3. The system needs to be configured as RAID 6. Given the block of data in the Figure 4 below, design them to allow dual redundancy.

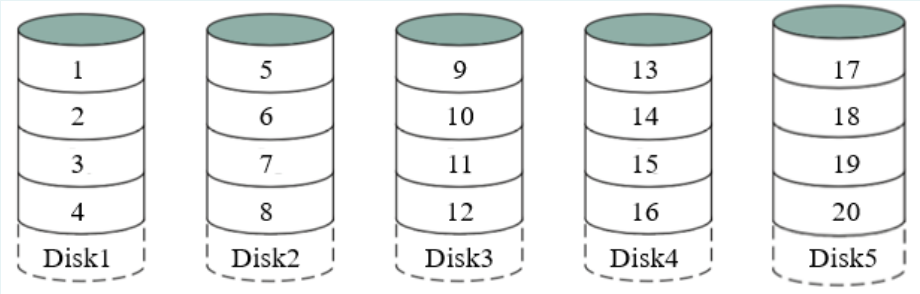


Figure 3: System with 5 disk drive

The number in the disks are the location. E.g. number 5 is referring to location 5

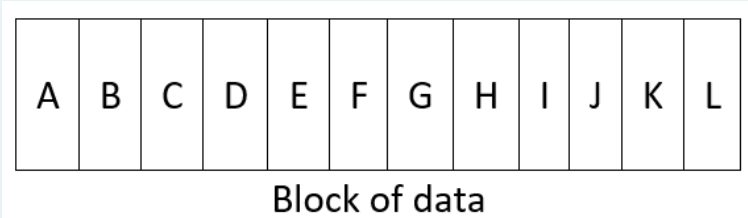


Figure 4: Block of data

a) Determine which block of data are allocated at location 3, 15 and 17
Answer format: Write only the letter with no white space. E.g. at location 1, A is written to represent data block A. Use uppercase letter P to represent parity block.

Location 3 =

Location 15 =

Location 17 =

b) If Disk1 and Disk4 failed, determine which location(s) should be used to regenerate the data at location 2.

Answer format: Write only the letter with no white space. If your answer has more than one location separate the letters with a comma e.g. location 1 and location 2 are written as 1,2 with a comma with no white space

Previous activity

◀ Final Exam Part A

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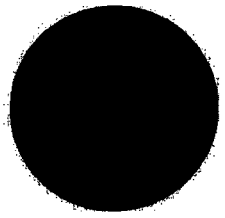
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FINAL EXAMINATION SEMESTER I , 2017 / 2018

SUBJECT CODE	:	**SCSR 2043 / SCSS 4213**
SUBJECT NAME	:	**OPERATING SYSTEMS**
SECTION	:	**ALL**
TIME	:	**9. 00 AM – 11 . 30 AM**
DATE/DAY	:	**08 / 01 / 2018 (MONDAY)**
VENUES	:	**BK 2 (N 28)**

INSTRUCTIONS :

Answer **ALL** questions in **section A** and **section B** in the answer booklet given.

(Please Write Your Lecture Name And Section In Your Answer Booklet)

Name	
I/C No.	
Year / Course	
Section	
Lecturer Name	

This questions paper consists of **ELEVEN (11)** printed pages excluding this page.

SECTION A

MULTIPLE CHOICE QUESTIONS (20 MARKS)

(INSTRUCTION: **Write all your answers** in the answer booklet provided)

1. Which of the following statements is **FALSE** regarding deadlock states?

   A. A deadlock state is an unsafe state.
   B. An unsafe state is a deadlock state.
   C. An unsafe state is not a deadlock state.
   D. Not all unsafe states is a deadlock. However, an unsafe state may cause a deadlock.

2. Which of the following statements is **TRUE** regarding deadlocks?

   A. Several processes may deadlock for an infinite number of resources.
   B. Deadlock can be avoided by allowing all four conditions to hold.
   C. Deadlock can lead to starvation.
   D. Deadlock can happened to independent processes.

3. _____ is a contiguous memory partitioning scheme, while _____ is a non-contiguous memory partitioning scheme.

   A. Segmentation, dynamic partition
   B. Dynamic partition, relocatable dynamic partition
   C. Paging, relocatable dynamic partition
   D. Relocatable dynamic partition, segmentation

4. The following statements is about Compaction, **EXCEPT**:

   A. Merge or combine the adjacent holes.
   B. Reduce internal fragmentation.
   C. All process can be moved to the one end of memory.
   D. Possible only if relocation is dynamic.

5. Which of the following statements about Paging is **FALSE**?

   A. In Paging, program pages will be loaded into frames.
   B. Internal fragmentation may still occurred in the last page.
   C. If the Page Size is too large, excessive internal fragmentation may happened.
   D. A Page Table is needed to do mapping of physical address to logical address.

6. Which of the following statements regarding Segmented Memory Allocation is **FALSE**?

   A. External fragmentation may occur on the last segment.

   B. Programs are allocated dynamically.

   C. Internal fragmentation is eliminated finallly.

   D. Segments are of different sizes.

7. The following phrases or statements are **TRUE** for paging **EXCEPT**:

   A. Main memory is allocated dynamically.

   B. Difficulty in managing page faults.

   C. Needs segment table.

   D. Equal page size.

8. In virtual memory demand-paging table, there is a column called valid-invalid bit.

   What is the purpose of this valid-invalid bit?

   A. To indicate whether the currently being accessed page need to be updated.

   B. To indicate whether the swapper is in error state or not.

   C. To indicate whether the currently being accessed page is currently in the memory or not.

   D. To indicate the process is invalid to be executed due to limited memory size that available for the process execution.

9. With virtual memory, size of logical memory of a job _____.

   A. must be less than the size of the physical memory

   B. may be greater than the size of the physical memory

   C. must be equal to the size of the secondary storage

   D. may be greater than the size of the secondary storage

10. Which of the following is **NOT** a reason to implement non-contiguous memory allocation scheme?

    A. Increase the complexity of the operating systems memory management.

    B. Elimination of compaction scheme.

    C. Avoidance of external fragmentation between page frames.

    D. Efficient usage of main memory.

11. Files are stored on the following devices or mediums **EXCEPT**:

A. primary storage and databases
B. secondary and tertiary storages
C. magnetic and optical storages
D. disks and tapes

12. There could be many external fragmentation and difficulty in file expansion occurred in the _____ allocation method.

A. Contiguous
B. Non-contiguous
C. Multi-level Indexed
D. Linked

13. The following statements most probably try to describe _____.

✓ *Inward sweep stops at last high-numbered track request.*
✓ *No last track access unless required.*

A. FCFS
B. C-Look
C. SSTF
D. C-Scan

14. Two common and main issues of using RAID for servers are _____.

A. performance and implementation issues
B. performance and redundancy issues
C. performance and reliability issues
D. performance and security issues

15. Disk performance or disk access time is normally based on the following **EXCEPT**:

A. seek time
B. waiting
C. transfer
D. latency

16. In Least Recently Used (LRU) page replacement algorithm, when a page must be replaced, _____.

A. last page is chosen
B. newest page is chosen
C. random page is chosen
D. oldest page accessed is chosen

17. Which of the following specifications describe RAID 5 clearly?

A. It requires one disk for redundancy.
B. Upon failure of a single drive, subsequent reads can be calculated from the strip parity such that no data is lost.
C. It consists of block-level striping with distributed parity.
D. It has designated parity disk.

18. In a tree-structured directory, the series of directory names that culminates in a file name is referred to as the _____.

A. Current Working Directory
B. File-system
C. Pathname
D. Symbolic name

19. The following allocation methods is best suited for files **EXCEPT**:

A. Contiguous—if file is usually accessed sequentially, and if file is relatively big.
B. Linked—if file is large and usually accessed sequentially.
C. Indexed—if file is large and usually accessed randomly.
D. Linked—if file is small and usually accessed sequentially.

20. These techniques can be used in managing free disk space **EXCEPT**:

A. Linked List
B. Swap space
C. Bitmaps
D. File Allocation Table

SECTION B

STRUCTURED QUESTIONS (80 MARKS)

(**INSTRUCTION**: **Write all your answers** in the answer booklet provided)

QUESTION 1 [15 MARKS]

a) Figure 1 shows 3 processes {P1,P2,P3} with 3 types of resources {R1,R2,R3}. Does the RAG below exhibit a deadlock? Justify your answer. [5 marks]

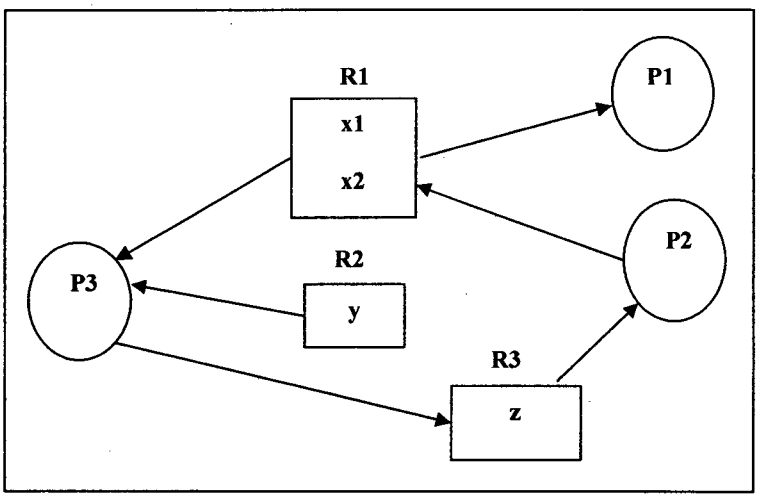


Figure 1 : Resource Allocation Graph

b) Given the following resource allocation and availability in Table 1.

Table 1

	Allocation			Maximum			Available		
	A	B	C	A	B	C	A	B	C
PI	2	2	1	6	2	6	3	2	4
P2	0	1	1	8	4	8			
P3	2	1	0	7	4	6			
P4	1	0	1	2	0	3			
P5	1	0	1	8	3	7			

i. Compute the Need Matrix based on the Safety Algorithm.
Note: Refer Appendix 1 for details of the algorithm. [2 marks]

ii. Compute the total number of resources in the system? [1 mark]

iii. Is the system in a safe state? Justify your answer. [5 marks]

iv. Suppose P2 requests (0, 2, 0), determine by using the Safety algorithm whether the operating system should grant the request by P2.
Show all your working. [2 marks]

QUESTION 2 [18 MARKS]

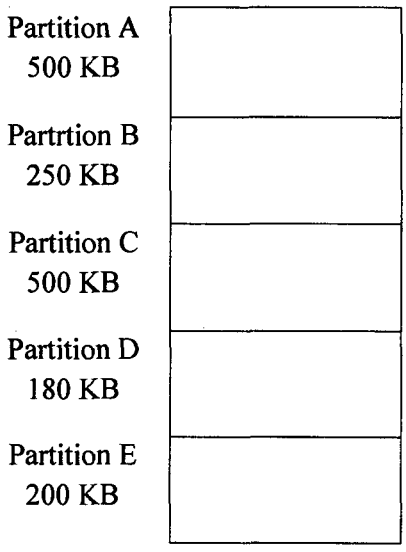


Figure 2: Memory partitions of Computer A

a) Total memory size of Computer A is 1630 KB. The memory has been partitioned as shown in Figure 2. The operating system implements fixed memory partitioning scheme. Assuming the memory partitions are still empty, and there are 5 processing jobs to be located into the memory. The jobs are as in the following:

Job 1 (300 KB), Job 2 (100KB), Job3 (220KB), Job 4 (500KB), Job 5 (180 KB).

Assuming jobs with lower ID number were created earlier than jobs with higher ID number. Answer below questions:

i. Assuming the operating system is using Best Fit memory allocation scheme. Sketch a figure to show the memory partition with all jobs are successfully being located into the memory. [3 marks]

ii. Compute the **size of internal fragmentation** and the **size of external fragmentation.** [2 marks]

b) Assuming there is a new computer that is called as Computer B. Main memory size of computer B is also 1630 KB. The operating system implements Relocatable Dynamic memory Partitioning scheme with Compaction. Sequence of jobs to be located into the memory is described as the following:

- In total, there are SEVEN (7) jobs to be processed by the computer.
- Jobs with lower ID number were created earlier than jobs with higher ID number.
- Size of Job 1-5 is exactly the same as Job 1-5 in (1).
- Size of Job 6 is 500 KB, and size of Job 7 is 50 KB.
- By the time Job 6 is to be located into the memory, Job 2 and Job 5 are already finished.

i. Sketch a figure to show the memory partition with only Job 1, 2, 3, 4 and 5 are in the memory. [3 marks]

ii. Sketch a figure to show the memory partition when Job 6 and Job 7 are already in the memory. [3 marks]

iii. Compute the **size of internal fragmentation** and the **size of external fragmentation.** [2 marks]

c) In an operating system, assume memory is allocated as specified in Figure 3.

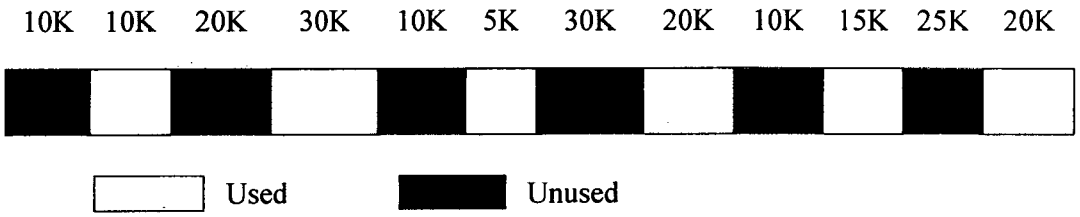


Figure 3: Existing memory allocation.

i. Redraw how the following set of jobs
{J1 – 30KB, J2 – 20KB, J3 – 15KB, J4 – 5KB} (in that order)
are located inside the memory using **next-fit allocation.** [2.5 Marks]

ii. Redraw how the following set of jobs
{J1 – 15KB, J2 - 10KB, J3 – 5KB, J4 – 30KB} (in that order)
are located inside memory using **best-fit allocation.** [2.5 marks]

QUESTION 3 [17 marks]

a) Explain the concept of Virtual Memory. In your explanation, you should also highlight several advantages of Virtual Memory, and give reasons why it is important and relevant in modern operating systems. [4 marks]

b) Given the following reference string:

1 3 2 1 4 3 2 4 1 5 2 4 2

For each of the following page replacement algorithms, calculate how many **page faults** will occur if **3 frames** are available:

i. Least Recently Used (LRU) [3 marks]

ii. Optimal Page Replacement (OPT) [3 marks]

iii. Based on your answer in a i) and a ii),
which algorithm is better? Justify your answer. [2 marks]

c) Answer the following questions about Trashing.

i. Define thrashing in the context of virtual memory. [2 marks]

ii. List and briefly describe TWO (2) techniques to prevent thrashing. [3 marks]

QUESTION 4 [15 MARKS]

a) On a simple paging system with 2^{28} bytes of physical memory, 512 pages of logical address space, and a page size of 2^{11} bytes, [7 marks]

i. How many bits are in a logical address?

ii. How many bytes are in a page frame?

iii. How many bits in the physical address specify the page frame?

iv. How many entries are in the page table?

v. What is the size of the physical memory (specify using the correct unit size)?

b) What advantage does segmentation offer over multiple variable partition? [2 marks]

c) Table 2 shows a simple segmentation system approach/method.

Table 2

Segment	Base	Length
0	350	124
1	860	308
2	104	144
3	480	120
4	600	40

For each of the following logical addresses, compute the corresponding physical addresses. If the adresss generates a segment fault, indicate so. [6 marks]

i. 2,104

ii. 3,120

iii. Is physical address 475 part of the program?

iv. What is the size of this program in bytes (Note: one address, 1Byte)?

QUESTION 5 [15 MARKS]

a) Figure 4 shows a diagram of disk storage containing two files, `File1` and `File2`:

Directory

File	Address	Size	Next
	1	512	
File1 (1)	2	512	8
File2 (1)	3	512	7
	. .	. .	
File2 (3)	6	512	-
File2 (2)	7	512	6
File1 (2)	8	512	20
	. .	. .	
File1 (4)	11	512	-
	. .	. .	
File1 (3)	20	512	11

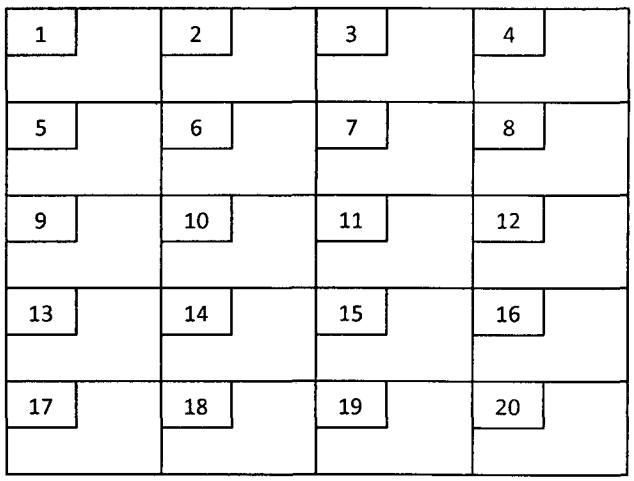


Figure 4: Disk Storage

i. Add arrows in the diagram according to the directory information of `File1` and `File2`. [3 marks]

ii. State the name of disk allocation technique that is being used. [2 marks]

b) On a disk with 600 tracks: numbered 0 to 599. The disk queue in FIFO order contains requests for the following tracks:

550, 42, 54, 294, 575, 208, 105

Starting from the current head location **134**, sketch figure to illustrate the head movement (track travelled) to serve all the requests based on

i. **FCFS** seek strategy. [2 marks]

ii. **SCAN** strategy. The head movement direction is inner-ward. [3 marks]

iii. **C-LOOK** strategy. The head movement direction is inner-ward. [3 marks]

iv. Based on the results of the above strategies, which seek strategy is the best? Justify your answer. [2 marks]

Appendix 1

Safety Algorithm

1. Let ***Work*** and ***Finish*** be vectors of length ***m*** and ***n***, respectively. Initialize:

   Work = Available

   Finish[i] = false for $i = 1, 2, \ldots, n$

2. Find an index ***i*** such that both

   Finish[i] == false

   $Need_i \leq Work$

   If no such ***i*** exists, go step 4.

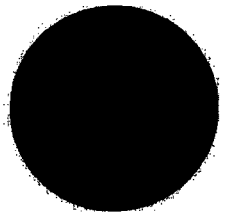
3. $Work = Work + Allocation_i$

   Finish [i] = true

   Goto step 2

4. If ***Finish[i] == true*** for all ***i*** → then the system in safe state

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FINAL EXAMINATION SEMESTER I , 2017 / 2018

SUBJECT CODE	:	**SCSR 2043 / SCSS 4213**
SUBJECT NAME	:	**OPERATING SYSTEMS**
SECTION	:	**ALL**
TIME	:	**9. 00 AM – 11 . 30 AM**
DATE/DAY	:	**08 / 01 / 2018 (MONDAY)**
VENUES	:	**BK 2 (N 28)**

INSTRUCTIONS :

Answer **ALL** questions in **section A** and **section B** in the answer booklet given.

(Please Write Your Lecture Name And Section In Your Answer Booklet)

Name	
I/C No.	
Year / Course	
Section	
Lecturer Name	

This questions paper consists of **ELEVEN (11)** printed pages excluding this page.

SECTION A

MULTIPLE CHOICE QUESTIONS (20 MARKS)

(INSTRUCTION: **Write all your answers** in the answer booklet provided)

1. Which of the following statements is **FALSE** regarding deadlock states?

   A. A deadlock state is an unsafe state.
   B. An unsafe state is a deadlock state.
   C. An unsafe state is not a deadlock state.
   D. Not all unsafe states is a deadlock. However, an unsafe state may cause a deadlock.

2. Which of the following statements is **TRUE** regarding deadlocks?

   A. Several processes may deadlock for an infinite number of resources.
   B. Deadlock can be avoided by allowing all four conditions to hold.
   C. Deadlock can lead to starvation.
   D. Deadlock can happened to independent processes.

3. _____ is a contiguous memory partitioning scheme, while _____ is a non-contiguous memory partitioning scheme.

   A. Segmentation, dynamic partition
   B. Dynamic partition, relocatable dynamic partition
   C. Paging, relocatable dynamic partition
   D. Relocatable dynamic partition, segmentation

4. The following statements is about Compaction, **EXCEPT**:

   A. Merge or combine the adjacent holes.
   B. Reduce internal fragmentation.
   C. All process can be moved to the one end of memory.
   D. Possible only if relocation is dynamic.

5. Which of the following statements about Paging is **FALSE**?

   A. In Paging, program pages will be loaded into frames.
   B. Internal fragmentation may still occurred in the last page.
   C. If the Page Size is too large, excessive internal fragmentation may happened.
   D. A Page Table is needed to do mapping of physical address to logical address.

6. Which of the following statements regarding Segmented Memory Allocation is **FALSE**?

   A. External fragmentation may occur on the last segment.

   B. Programs are allocated dynamically.

   C. Internal fragmentation is eliminated finallly.

   D. Segments are of different sizes.

7. The following phrases or statements are **TRUE** for paging **EXCEPT**:

   A. Main memory is allocated dynamically.

   B. Difficulty in managing page faults.

   C. Needs segment table.

   D. Equal page size.

8. In virtual memory demand-paging table, there is a column called valid-invalid bit.

   What is the purpose of this valid-invalid bit?

   A. To indicate whether the currently being accessed page need to be updated.

   B. To indicate whether the swapper is in error state or not.

   C. To indicate whether the currently being accessed page is currently in the memory or not.

   D. To indicate the process is invalid to be executed due to limited memory size that available for the process execution.

9. With virtual memory, size of logical memory of a job _____.

   A. must be less than the size of the physical memory

   B. may be greater than the size of the physical memory

   C. must be equal to the size of the secondary storage

   D. may be greater than the size of the secondary storage

10. Which of the following is **NOT** a reason to implement non-contiguous memory allocation scheme?

    A. Increase the complexity of the operating systems memory management.

    B. Elimination of compaction scheme.

    C. Avoidance of external fragmentation between page frames.

    D. Efficient usage of main memory.

11. Files are stored on the following devices or mediums **EXCEPT**:

A. primary storage and databases
B. secondary and tertiary storages
C. magnetic and optical storages
D. disks and tapes

12. There could be many external fragmentation and difficulty in file expansion occurred in the _____ allocation method.

A. Contiguous
B. Non-contiguous
C. Multi-level Indexed
D. Linked

13. The following statements most probably try to describe _____.

✓ *Inward sweep stops at last high-numbered track request.*
✓ *No last track access unless required.*

A. FCFS
B. C-Look
C. SSTF
D. C-Scan

14. Two common and main issues of using RAID for servers are _____.

A. performance and implementation issues
B. performance and redundancy issues
C. performance and reliability issues
D. performance and security issues

15. Disk performance or disk access time is normally based on the following **EXCEPT**:

A. seek time
B. waiting
C. transfer
D. latency

16. In Least Recently Used (LRU) page replacement algorithm, when a page must be replaced, _____.

A. last page is chosen
B. newest page is chosen
C. random page is chosen
D. oldest page accessed is chosen

17. Which of the following specifications describe RAID 5 clearly?

A. It requires one disk for redundancy.
B. Upon failure of a single drive, subsequent reads can be calculated from the strip parity such that no data is lost.
C. It consists of block-level striping with distributed parity.
D. It has designated parity disk.

18. In a tree-structured directory, the series of directory names that culminates in a file name is referred to as the _____.

A. Current Working Directory
B. File-system
C. Pathname
D. Symbolic name

19. The following allocation methods is best suited for files **EXCEPT**:

A. Contiguous—if file is usually accessed sequentially, and if file is relatively big.
B. Linked—if file is large and usually accessed sequentially.
C. Indexed—if file is large and usually accessed randomly.
D. Linked—if file is small and usually accessed sequentially.

20. These techniques can be used in managing free disk space **EXCEPT**:

A. Linked List
B. Swap space
C. Bitmaps
D. File Allocation Table

SECTION B

STRUCTURED QUESTIONS (80 MARKS)

(**INSTRUCTION**: **Write all your answers** in the answer booklet provided)

QUESTION 1 [15 MARKS]

a) Figure 1 shows 3 processes {P1,P2,P3} with 3 types of resources {R1,R2,R3}. Does the RAG below exhibit a deadlock? Justify your answer. [5 marks]

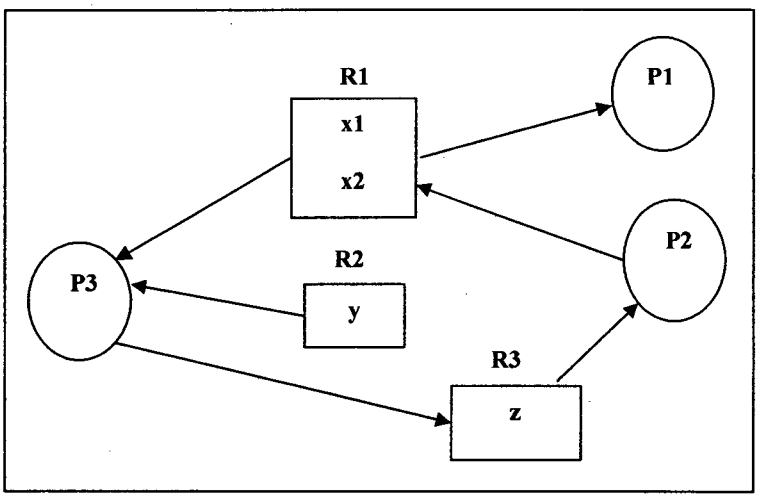


Figure 1 : Resource Allocation Graph

b) Given the following resource allocation and availability in Table 1.

Table 1

	Allocation			Maximum			Available		
	A	B	C	A	B	C	A	B	C
PI	2	2	1	6	2	6	3	2	4
P2	0	1	1	8	4	8			
P3	2	1	0	7	4	6			
P4	1	0	1	2	0	3			
P5	1	0	1	8	3	7			

i. Compute the Need Matrix based on the Safety Algorithm.
Note: Refer Appendix 1 for details of the algorithm. [2 marks]

ii. Compute the total number of resources in the system? [1 mark]

iii. Is the system in a safe state? Justify your answer. [5 marks]

iv. Suppose P2 requests (0, 2, 0), determine by using the Safety algorithm whether the operating system should grant the request by P2.
Show all your working. [2 marks]

QUESTION 2 [18 MARKS]

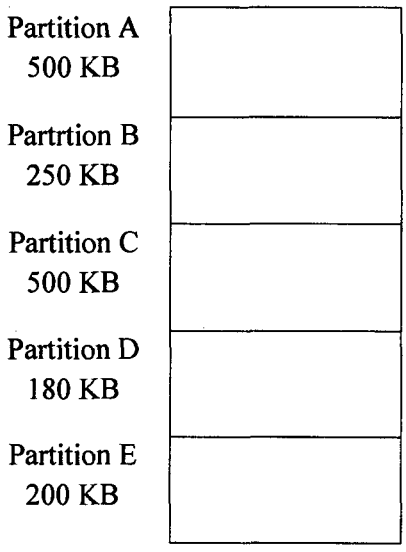


Figure 2: Memory partitions of Computer A

a) Total memory size of Computer A is 1630 KB. The memory has been partitioned as shown in Figure 2. The operating system implements fixed memory partitioning scheme. Assuming the memory partitions are still empty, and there are 5 processing jobs to be located into the memory. The jobs are as in the following:

Job 1 (300 KB), Job 2 (100KB), Job3 (220KB), Job 4 (500KB), Job 5 (180 KB).

Assuming jobs with lower ID number were created earlier than jobs with higher ID number. Answer below questions:

i. Assuming the operating system is using Best Fit memory allocation scheme. Sketch a figure to show the memory partition with all jobs are successfully being located into the memory. [3 marks]

ii. Compute the **size of internal fragmentation** and the **size of external fragmentation.** [2 marks]

b) Assuming there is a new computer that is called as Computer B. Main memory size of computer B is also 1630 KB. The operating system implements Relocatable Dynamic memory Partitioning scheme with Compaction. Sequence of jobs to be located into the memory is described as the following:

- In total, there are SEVEN (7) jobs to be processed by the computer.
- Jobs with lower ID number were created earlier than jobs with higher ID number.
- Size of Job 1-5 is exactly the same as Job 1-5 in (1).
- Size of Job 6 is 500 KB, and size of Job 7 is 50 KB.
- By the time Job 6 is to be located into the memory, Job 2 and Job 5 are already finished.

i. Sketch a figure to show the memory partition with only Job 1, 2, 3, 4 and 5 are in the memory. [3 marks]

ii. Sketch a figure to show the memory partition when Job 6 and Job 7 are already in the memory. [3 marks]

iii. Compute the **size of internal fragmentation** and the **size of external fragmentation.** [2 marks]

c) In an operating system, assume memory is allocated as specified in Figure 3.

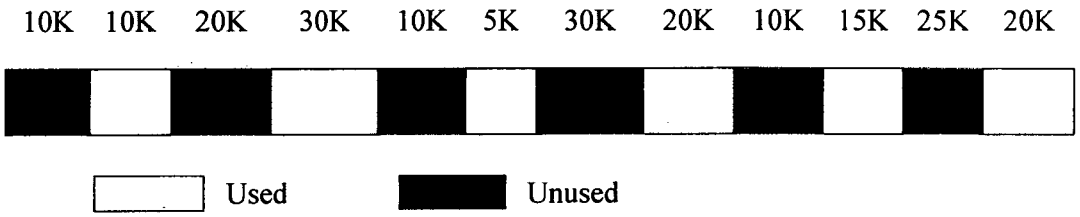


Figure 3: Existing memory allocation.

i. Redraw how the following set of jobs
{J1 – 30KB, J2 – 20KB, J3 – 15KB, J4 – 5KB} (in that order)
are located inside the memory using **next-fit allocation.** [2.5 Marks]

ii. Redraw how the following set of jobs
{J1 – 15KB, J2 - 10KB, J3 – 5KB, J4 – 30KB} (in that order)
are located inside memory using **best-fit allocation.** [2.5 marks]

QUESTION 3 [17 marks]

a) Explain the concept of Virtual Memory. In your explanation, you should also highlight several advantages of Virtual Memory, and give reasons why it is important and relevant in modern operating systems. [4 marks]

b) Given the following reference string:

1 3 2 1 4 3 2 4 1 5 2 4 2

For each of the following page replacement algorithms, calculate how many **page faults** will occur if **3 frames** are available:

i. Least Recently Used (LRU) [3 marks]

ii. Optimal Page Replacement (OPT) [3 marks]

iii. Based on your answer in a i) and a ii),
which algorithm is better? Justify your answer. [2 marks]

c) Answer the following questions about Trashing.

i. Define thrashing in the context of virtual memory. [2 marks]

ii. List and briefly describe TWO (2) techniques to prevent thrashing. [3 marks]

QUESTION 4 [15 MARKS]

a) On a simple paging system with 2^{28} bytes of physical memory, 512 pages of logical address space, and a page size of 2^{11} bytes, [7 marks]

i. How many bits are in a logical address?

ii. How many bytes are in a page frame?

iii. How many bits in the physical address specify the page frame?

iv. How many entries are in the page table?

v. What is the size of the physical memory (specify using the correct unit size)?

b) What advantage does segmentation offer over multiple variable partition? [2 marks]

c) Table 2 shows a simple segmentation system approach/method.

Table 2

Segment	Base	Length
0	350	124
1	860	308
2	104	144
3	480	120
4	600	40

For each of the following logical addresses, compute the corresponding physical addresses. If the adresss generates a segment fault, indicate so. [6 marks]

i. 2,104

ii. 3,120

iii. Is physical address 475 part of the program?

iv. What is the size of this program in bytes (Note: one address, 1Byte)?

QUESTION 5 [15 MARKS]

a) Figure 4 shows a diagram of disk storage containing two files, `File1` and `File2`:

Directory

File	Address	Size	Next
	1	512	
File1 (1)	2	512	8
File2 (1)	3	512	7
	. .	. .	
File2 (3)	6	512	-
File2 (2)	7	512	6
File1 (2)	8	512	20
	. .	. .	
File1 (4)	11	512	-
	. .	. .	
File1 (3)	20	512	11

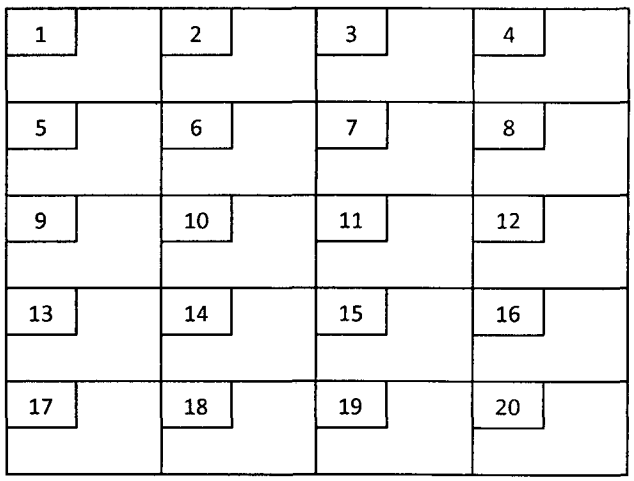


Figure 4: Disk Storage

i. Add arrows in the diagram according to the directory information of `File1` and `File2`. [3 marks]

ii. State the name of disk allocation technique that is being used. [2 marks]

b) On a disk with 600 tracks: numbered 0 to 599. The disk queue in FIFO order contains requests for the following tracks:

550, 42, 54, 294, 575, 208, 105

Starting from the current head location **134**, sketch figure to illustrate the head movement (track travelled) to serve all the requests based on

i. **FCFS** seek strategy. [2 marks]

ii. **SCAN** strategy. The head movement direction is inner-ward. [3 marks]

iii. **C-LOOK** strategy. The head movement direction is inner-ward. [3 marks]

iv. Based on the results of the above strategies, which seek strategy is the best? Justify your answer. [2 marks]

Appendix 1

Safety Algorithm

1. Let ***Work*** and ***Finish*** be vectors of length ***m*** and ***n***, respectively. Initialize:

   Work = Available

   Finish[i] = false for $i = 1, 2, \ldots, n$

2. Find an index ***i*** such that both

   Finish[i] == false

   $Need_i \leq Work$

   If no such ***i*** exists, go step 4.

3. $Work = Work + Allocation_i$

   Finish [i] = true

   Goto step 2

4. If ***Finish[i] == true*** for all ***i*** → then the system in safe state

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FINAL EXAMINATION SEMESTER II , 2017 / 2018

SUBJECT CODE	:	**SCSR 2043**
SUBJECT NAME	:	**OPERATING SYSTEMS**
SECTION	:	**ALL**
TIME	:	**9. 00 AM – 11 . 30 AM**
DATE/DAY	:	**01 / 06 / 2018 (FRIDAY)**
VENUES	:	**KTDI**

INSTRUCTIONS :

Answer **ALL** questions in the answer booklet given.

(Please Write Your Lecture Name And Section In Your Answer Booklet)

Name	
I/C No.	
Year / Course	
Section	
Lecturer Name	

This questions paper consists of **THREE (3)** printed pages excluding this page.

INSTRUCTION: Please answer all questions in the answer booklet. Show all your workings.

1. **[20 MARKS]** There are 5 processes (in order) of *A* (115 KB), *B* (500 KB), *C* (350 KB), *D* (200 KB), and *E* (450 KB) need to be placed in the memory.

   a) Suppose a fixed-partition is used with six memory partitions of 300 *KB*, 600 *KB*, 400 *KB*, 200 *KB*, 750 *KB*, and 125 *KB* (in order).
      (i) How would the best-fit and worst-fit algorithms place the processes? Draw the memory stack and label the partition size clearly for both algorithms. [10m]
      (ii) In terms of process placement in the memory, which algorithm faced a problem? What is the problem and state the cause of the problem. [4m]

   b) Suppose a variable or dynamic partition is used. The following events happened:
      t_1: process A and D left.
      t_2: process F (50 *KB*) enters
      t_3: process B left.
      t_4: process G (550 *KB*) enters.

      Draw the memory stack at each t_2, t_3 and t_4 and label the sizes clearly for all processes. Use first-fit algorithm and relocatable dynamic partitions of merging adjacent holes to place the process in the memory. [6m]

2. **[10 MARKS]** Consider a logical address space of 2^8 pages with a 4 *KB* page size, mapped onto a physical memory of 4096 frames. Its page table is given in Table 2.

Table 2

0	50
1	30
2	400
3	100

   a) What is the size of logical and physical addresses (in 2^n bytes)? [4m]
   b) Calculate the physical address if the logical address is 3:400. [3m]
   c) What is the page number and offset for the logical address of 51152 (in decimal)? [3m]

3. **[12 MARKS]** Define the following terms.
   a) Demand paging [3m]
   b) Lazy swapper [3m]
   c) Belady's algorithm [3m]
   d) Thrashing in virtual memory [3m]

4. **[13 MARKS]** Consider the following page reference string:

   7, 2, 3, 1, 2, 5, 3, 4, 6, 7, 1, 3

   Assuming demand paging with four frames, how many page faults would occur for the following replacement algorithms? Please circle which references cause page faults.
   a) Optimal (OPT) replacement [5m]
   b) LRU replacement [7m]

5. **[10 MARKS]** Draw and discuss the advantages of:
   a) Tree-structured directories [5m]
   b) Acyclic-graph directories [5m]

6. **[10 MARKS]** Answer the following questions:
   a) In the Unix system, directory listing is shown as below:

```
-rw-r--r--      1 pbg staff      9423    Apr 19 2018   program
drwx------      3 pbg staff      1024    Aug 19 10:31  mail/
```

      (i) Name and define the directory protection. [3m]
      (ii) Name and define the file protection. [3m]
      (iii) How to set all the file protection using chmod? [4m]

7. **[10 MARKS]** Describe and show the example of free space management using:
   a) Linked list [5m]
   b) Grouping [5m]

8. **[15 MARKS]** Consider the following information:

- Suppose that a disk drive has 5000 cylinders, numbered 0 – 4999.
- Using C-SCAN disk-scheduling algorithm to satisfy all the cylinder requests.
- The drive is currently serving a request at cylinder 2150.
- The queue of pending requests is:

2069, 1212, 2296, 2800, 544, 1618, 356, 1523, 4965, 3681

Starting from current head position, answer the following questions.

a) Draw the whole disk arm movement to satisfy all the pending requests. [7m]

b) What is the total seek distance (in cylinder) that the disk arm had moved? [3m]

c) Now, if C-LOOK algorithm is used to satisfy all the cylinder pending requests, compare the efficiency of both disk-scheduling algorithms based on the distance (in cylinder) that the disk arm moved. [5m]

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FINAL EXAMINATION SEMESTER II 2021/2022

SUBJECT CODE	**: SECR2043**
SUBJECT NAME	**: OPERATING SYSTEM**
SECTION	**: SECR/J/B/V/P/H / SCSR/J/B/V/P/H**
TIME	**: PAPER 1 – 90 MINUTES**
DATE/DAY	**: 18 JULY 2022 (MONDAY)**
VENUES	**: Online**

INSTRUCTIONS:

Answer all questions from Paper 1 and Paper 2. Show all your works.

This test will contribute 35% towards the total marks of 100%.

(Please Write Your Lecturer Name and Section in Your Answer Booklet)

Name	
I/C No.	
Year / Course	
Section (Circle)	**01 / 02 / 03 / 04 / 05 / 06 / 07 / 08 / 09 / 15**
Lecturer Name (Circle)	**Dr Zuriahati / Mrs Rashidah** **Dr Nur Haliza / Dr Nor Shahida**

This question paper consists of 4 printed pages excluding this page.

QUESTION 1 [25 MARKS])

a) Refer to the following set of edges that consists of request edges (R) and assignment edges (P). Answer the following questions.

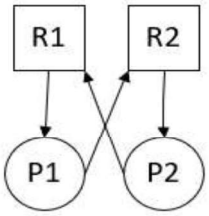
$$E = \{P_1 \rightarrow R_1, R_1 \rightarrow P_2, R_1 \rightarrow P_5, P_5 \rightarrow R_2, R_2 \rightarrow P_3, P_3 \rightarrow R_3, R_3 \rightarrow P_2, R_3 \rightarrow P_1, R_3 \rightarrow P_4\}$$

i. By assuming $R_1, R_2,$ **and** R_3 has 3, 3 and 1 instance/s respectively. Draw a complete Resource-Allocation Graph (RAG). Label all components clearly including the symbol ' • ' to represent an instance/s. [4]

ii. Based on the RAG from Question 1(a), state whether the system is in a deadlock state or not. Justify your answer and circle (if any) on your RAG from Question 1(a) to denote the deadlock occurrence. [2]

iii. If P_4, has released the R_3, what happened to the deadlock state of the system? Justify your answer your answer. [2]

b) Refer to the following RAG, construct its corresponding **wait-for** graph to detect whether the system is in a deadlock state or not. State your conclusion? [3]



c) Refer to Table 1 and answer the following questions. Consider the Available = $(3, 3, 2, 1)$.

Table 1

	Allocation				Maximum			
	A	***B***	***C***	***D***	***A***	***B***	***C***	***D***
P_0	2	0	1	0	5	0	5	2
P_1	2	1	0	1	2	3	1	5
P_2	3	2	1	2	5	3	2	3
P_3	0	3	2	1	2	5	5	4

i. By employing Resource Request Algorithm, identify whether a request from P_3 that arrives for (1, 2, 0, 0) can be granted immediately or not? If yes, write the safe sequence. Show all your work. [12]

ii. How many totals of instances for each of the resource A, B, C and D? [2]

QUESTION 2 [25 MARKS])

a) A paging system with 512 pages of logical address space, a page size of 2^8 and number of frames is 1024. Answer the following questions (in bytes or in form 2^n) and show all your work.

Page number	Frame number
0	100
1	50
2	528
3	10
4	79
5	644
6	21
7	301

i. What is the total size of the logical address space? [2]

ii. What is the total size of the physical address space? [2]

iii. What is the frame size of the physical memory? [1]

iv. How many frames of the physical memory in a number of the form 2^m? [1]

v. How many number of entry in the page table? [1]

vi. What is the number of bit for the page number in logical memory address and frame number in physical memory address? [2]

vii. What is the logical address (in decimal) given a physical address of 100001000010_2. [2]

b) There are 4 jobs (in order) of *A* (100 KB), *B* (350 KB), *C* (450 KB) and *D* (220 KB) need to be placed in the memory. Assume a fixed partition is used with six memory partitions of 200 *KB*, 300 *KB*, 125 *KB*, 600 *KB*, 450 *KB* and 250 *KB* (in order).

i. Use the Best-fit algorithm to indicate which memory blocks are allocated to each of the four arriving jobs. Draw the memory stack and label the partition size clearly (shaded for unused). [4]

ii. Does internal fragmentation occur? If yes, calculate the size of internal fragmentation before merging? [2]

c) Assume a dynamic or variable partition is used. The following events happened:

t_1: process A and D left.

t_2: process F (550 *KB*) enters.

Based on **question 2b(i) answer**, draw and update the event of the memory stack at each t_1 and t_2, and label the sizes clearly for all jobs. Use the **Best-fit algorithm and relocatable dynamic partitions** of merging the holes to place the process in the memory (shaded for unused). [8]

i. t_0: _____ and t_1: _____

ii. t_1: _____

iii. t_1: _____

iv. t_2: _____

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FINAL EXAMINATION SEMESTER II 2021/2022

SUBJECT CODE	**: SECR2043**
SUBJECT NAME	**: OPERATING SYSTEM**
SECTION	**: SECR/J/B/V/P/H/ SCSR/J/B/V/P/H**
TIME	**: PAPER 2 – 90 MINUTES**
DATE/DAY	**: 18 JULY 2022 (MONDAY)**
VENUES	**: Online**

INSTRUCTIONS:

Answer all questions from Paper 1 and Paper 2. Show all your works.

This test will contribute 35% towards the total marks of 100%.

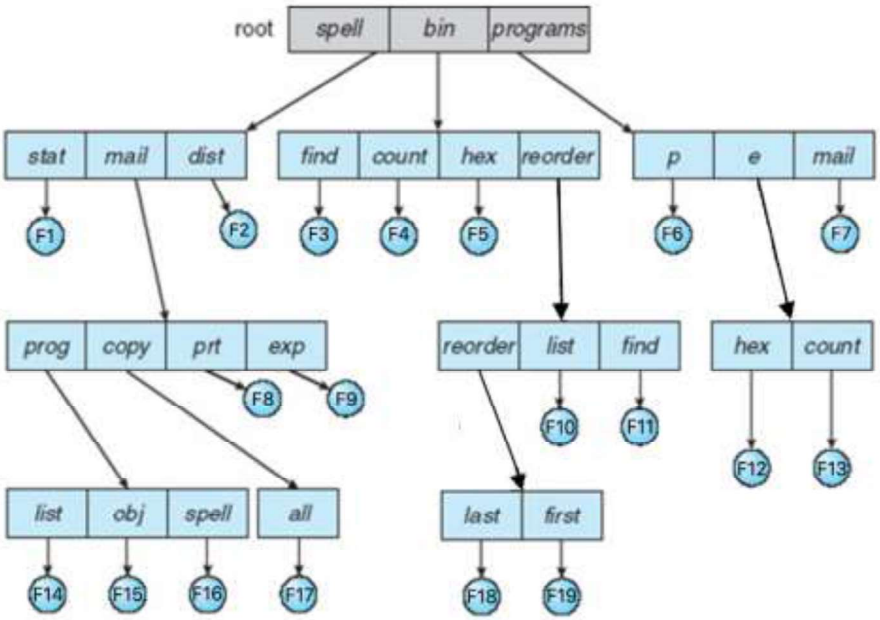
(Please Write Your Lecturer Name and Section in Your Answer Booklet)

Name	
I/C No.	
Year / Course	
Section (Circle)	**01 / 02 / 03 / 04 / 05 / 06 / 07 / 08 / 09 / 15**
Lecturer Name (Circle)	**Dr Zuriahati / Mrs Rashidah** **Dr Nur Haliza / Dr Nor Shahida**

This question paper consists of 4 printed pages excluding this page.

QUESTION 3 [15 MARKS])

a) Refer to the diagram below, assume the `LINUX` command prompt environment is used and answer the following questions. Show all your work.



i. Determine the absolute path name for file, `F6` and `F19`. [2]

ii. Determine the relative path name for file, `F7` and `F17`. [2]

iii. Identify the current directory for file, `F8`. [1]

iv. A filename without the path information can also be called as: [1]

v. File `F16` has a subdirectory with 9 protection bits, 3 for owner, 3 for group and 3 for public. Each class is assigned with **R**, **W** and **D**. Following the value (in decimal) assigned to the class, explain the access permission for each of the classes. [3]

`Owner = 6, group = 5, public = 1`

vi. File `F16` later changed using this command "`chmod 743 F16` " (for access read, write, execute), write the `UNIX` directory for the file `F16`. [2]

vii. Explain briefly what is the explanation for the `UNIX` directory given. [2]

```
-rwxrwxr-x 7 research admin 512 Jul 16 01.12 public.doc
drwx-wxr-x 2 research librarian 1024 Aug 31 03.23 public/
```

viii. In the new directory, write **TWO** possible single command lines to set an access control of the `.doc` file to be classes as read only. [2]

QUESTION 4 [20 MARKS])

Consider the reference string:

4	3	2	1	2	3	4	7	3	2	1	2	4	5	1	2	1	3	2	1

a) How many page faults if we have 4 frames using FIFO? Show all your work. [4]

b) How many page faults if we have 3 frames using FIFO? Show all your work. [4]

c) From the results above, perform the comparative analysis and discuss about system efficiency related to the page faults amount and the frame number used for FIFO. [4]

d) How many page faults if we have 4 frames using OPT? [4]

e) From the results above, perform the comparative analysis and discuss about system efficiency related the page faults that related with the different algorithms used. [4]

QUESTION 5 [15 MARKS])

a) By using the C-SCAN disk scheduling algorithms, illustrated by drawing the movement of the disk arm to satisfy the requests. Label clearly. [5]

- ✓ A disk with 160 cylinders.
- ✓ A request queue:
  97, 100, 50, 134, 26, 145, 80, 90
- ✓ Head pointer 75.
- ✓ Direction: the disk arm is moving to innermost

b) Calculate the total distance (in cylinders) that the disk arm moves to satisfy the requests. Show all your work. [2]

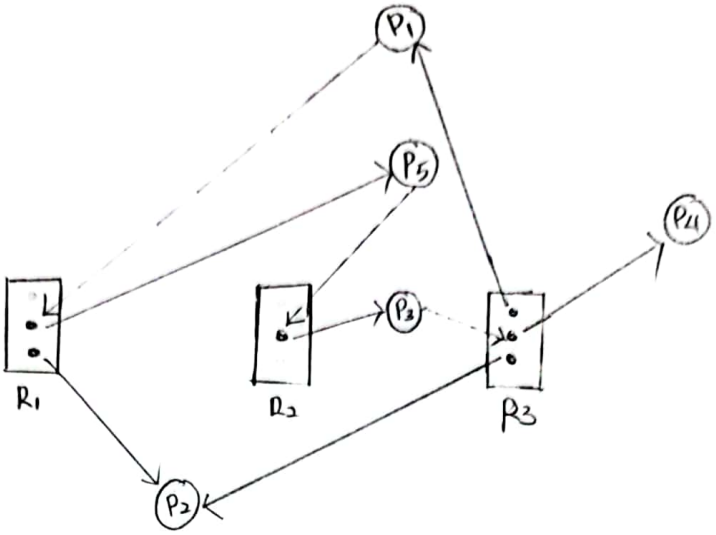
c) If the SCAN disk scheduling algorithm is used, briefly state a conclusion that can be made about the disk arm movement to satisfy the requests. [3]

d) Using similar details from question (a), redraw the movement of the disk arm using SCAN algorithm with direction of the disk arm moving toward 0. Label clearly. [5]

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Question 1

a) i)



ii) The RAG has one cycle:

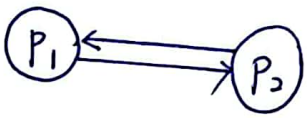
$P_1 \rightarrow R_1 \rightarrow P_5 \rightarrow R_2 \rightarrow P_3 \rightarrow R_3 \rightarrow P_1$

iii) P_1, P_3, P_5 are in a deadlock state.

∴ this system is not deadlock

when P_4 releases R_3, this resource can be assigned to P_3.

b)



A cycle from P1 to P2 and again to P1.

The system is in a deadlock state.

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c) i) Allocation

	A	B	C	D
P_0	2	0	1	0
P_1	2	1	0	1
P_2	3	2	1	2
P_3	0	3	2	1

Max

A	B	C	D
5	0	5	2
2	3	1	5
5	3	2	3
2	5	5	4

Need

A	B	C	D
3	0	4	2
0	2	1	4
2	1	1	1
2	2	3	3

P_3 Request $(1,2,0,0) \leq$ Available $(3,3,2,1)$

Request$_3 \leq$ Need$_3$

Available $= (3,3,2,1) - (1,2,0,0)$
$= (2,1,2,1)$

Allocation $(P_3) = (0,3,2,1) + (1,2,0,0)$
$= (1,5,2,1)$

Need $(P_3) = (2,2,3,3) + (1,2,0,0)$
$= (3,4,3,3)$

Allocation

	A	B	C	D
P_0	2	0	1	0
P_1	2	1	0	1
P_2	3	2	1	2
P_3	1	5	2	1

Max

A	B	C	D
5	0	5	2
2	3	1	5
5	3	2	3
2	5	5	4

Need.

A	B	C	D
3	0	4	2
0	2	1	4
2	1	1	1
3	4	3	3

1. Work = Available $(2,1,2,1)$
2. Need $(P_2) \leq$ Work
   $(2,1,1,1) \leq (2,1,2,1)$
   Work $= (2,1,2,1) + (3,2,1,2)$
   $= (5,3,3,3)$
3. Need $(P_1) \leq$ Work
   $(0,2,1,4) \leq (5,3,3,3)$
   Work $= (5,3,3,3) + (2,1,0,1)$
   $= (7,4,3,4)$
4. Need $(P_3) \leq$ Work
   $(3,4,3,3) \leq (7,4,3,4)$
   Work $= (7,4,3,4) + (1,5,2,1)$
   $= (8,9,5,5)$
5. Need $(P_0) \leq$ Work
   $(3,0,4,2) \leq (8,9,5,5)$
   Work $= (8,9,5,5) + (2,0,1,0)$
   $= (10,9,6,5)$

Safe sequence

$P_2 \rightarrow P_1 \rightarrow P_3 \rightarrow P_0$

ii) Total instances

A: $2+2+3+1=8$
B: $0+1+2+5=8$
C: $1+0+1+2=4$
D: $0+1+2+1=4$

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Question 2.

a)

Page size = frame size = 2^8

No. of frame = 1024

No. of page = 512 pages.

i) Total size of the logical address space
= total pages x page size
= 512 x 2^8
= 131072 bytes.

ii) Total size of the physical address space
= total frames x frame size
= 1024 x 2^8
= 262144 bytes.

iii) Page size = frame size = 2^8 = 256

iv.) No. of frames = 1024 = 2^{10}.

v.) Number of entry = number of pages
= 512 entries.

vi)

offset bit = 8 bits.

Logical memory
= 131072
= 2^{17}

Page number bit
= 17 - 8
= 9 bits.

Physical memory
= 262144
= 2^{18}

Frame number bit
= 18 - 8
= 10 bits

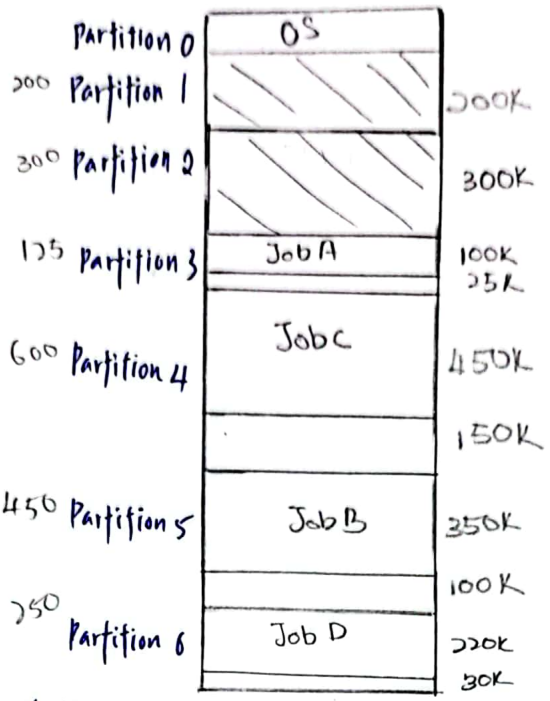
vii) Frame # : frame offset
1000 : 0100 0010 → 8 : 66

Logical address
= page # x page size + offset
= 3 x 2^8 + 66
= 834

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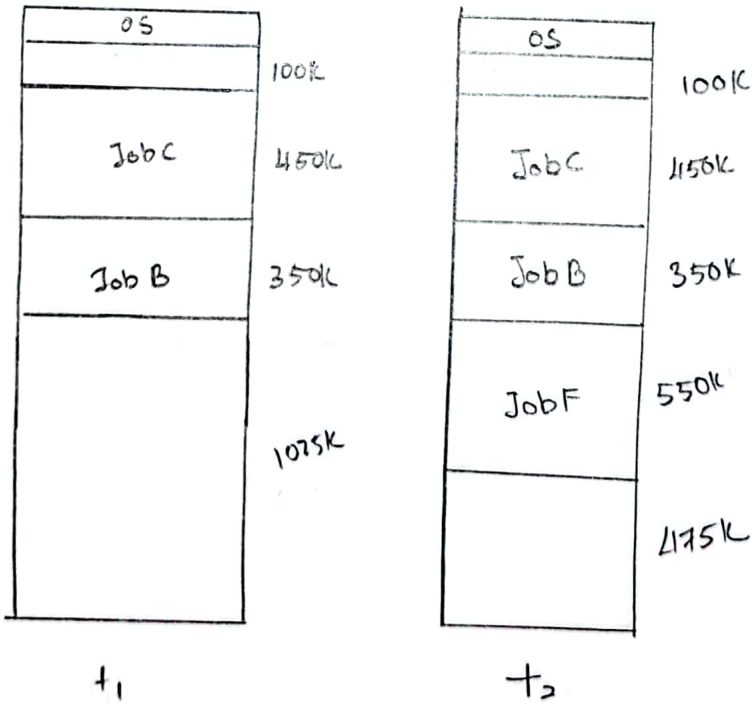
Question 2(b)

b) i) Jobs :
A (100KB)
B (350KB)
C (450KB)
D (220KB)



ii) Internal fragmentation occur.
Yes.
Size of internal fragmentation
= 25K + 150K + 100K + 30K
= 305K

c)



i) to : 1120K and t_1 = 800K
ii) t_1 = 800K
iii) t_1 = 800K
iv) t_2 = 1350K

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UNIVERSITI TEKNOLOGI MALAYSIA
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SUBJECT CODE	**: SECR2043**
SUBJECT NAME	**: OPERATING SYSTEM**
SECTION	**: SECR/J/B/V/P/H/ SCSR/J/B/V/P/H**
TIME	**: PAPER 2 – 90 MINUTES**
DATE/DAY	**: 18 JULY 2022 (MONDAY)**
VENUES	**: Online**

INSTRUCTIONS:

Answer all questions from Paper 1 and Paper 2. Show all your works.

This test will contribute 35% towards the total marks of 100%.

(Please Write Your Lecturer Name and Section in Your Answer Booklet)

Name	TEH JING LING
I/C No.	010111-08-0090
Year / Course	2/SECJH
Section (Circle)	**01 / 02 / 03 / 04 / 05 / 06 / 07 / 08 / 09 / 15**
Lecturer Name (Circle)	**Dr Zuriahati / Mrs Rashidah** **Dr Nur Haliza / Dr Nor Shahida**

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TEH JING LING A20EC0228

Question 3

a) i) root / programs / p / F6

root / bin / recorder / recorder / first / F19

ii) mail / F7

All / F17

iii) cd /spell / mail / prf

iv) Relative filename

v)

Owner	Group	public
6	5	1
110	101	001
RW-	R-D	--D

Owner allowed only read and write File F16.
Group allowed only read and delete File F16.
Public allowed only delete File F16.

vi) -rwxr---wx 7 research admin 512 Sep 3 08:30 F16

vii) "public.doc" is a filed owned by "research" with group "admin".
"public/" is a subdirectory owned by "research" with group "librarian".

viii) chmod 444 *.doc

chmod a+r, a-wx *doc.

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Question 4

a)

	4	3	2	1	2	3	4	7	3	2	1	2	4	5	1	2	1	3	2	1
Frame #0	4	4	4	4				7					7	7		7	1	1		
Frame #1		3	3	3				3					4	4		4	4	3		
Frame #2			2	2				2					2	5		5	5	5		
Frame #3				1				1					1	1		2	2	2		

No. of page fault = 10

b)

	4	3	2	1	2	3	4	7	3	2	1	2	4	5	1	2	1	3	2	1
Frame#0	4	4	4	1			1	1	3	3	3		4	4		4	1	1		
Frame#1		3	3	3			4	4	4	2	2		2	5		5	5	3		
Frame#2			2	2			2	7	7	7	1		1	1		2	2	2		

No. of page fault = 14

c) 4 frames have less 4 page faults than 3 frames by using FIFO. (14 - 10 = 4)

More frame number can reduce page fault amount.

The system efficiency of 4 frames is greater than system efficiency of 3 frames. Since the higher the frame number, the lower the page fault amount.

Belady's Anomoly

More frames may lead to more page faults.

d)

	4	3	2	1	2	3.	4	7	3	2	1	2	4	5	1	2	1	3	2	1
Frame#0	4	4	4	4				7					4	5						
Frame#1		3	3	3				3					3	3						
Frame#2			2	2				2					2	2						
Frame#3				1				1					1	1						

No. of page fault = 7

e) OPT have less 3 page faults than FIFO (10 - 7 = 3)

The system efficiency of OPT is greater than the system efficiency of FIFO since it has less page fault number compered to FIFO.

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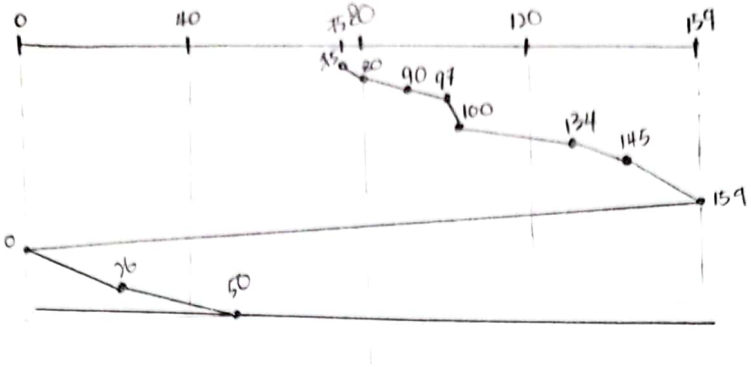
Question 5

5 6 2 7 1 4 3 8

97, 100, 50, 134, 26, 145, 80, 90

26, 50, 80, 90, 97, 100, 134, 145

a)



b) Total distance

= 159 - 75 + 159 - 0 + 50 - 0

= 84 + 159 + 50

= 293 cylinders.

c)

Total distance

= 159 - 75 + 159 - 26

= 217 cylinders.

Difference

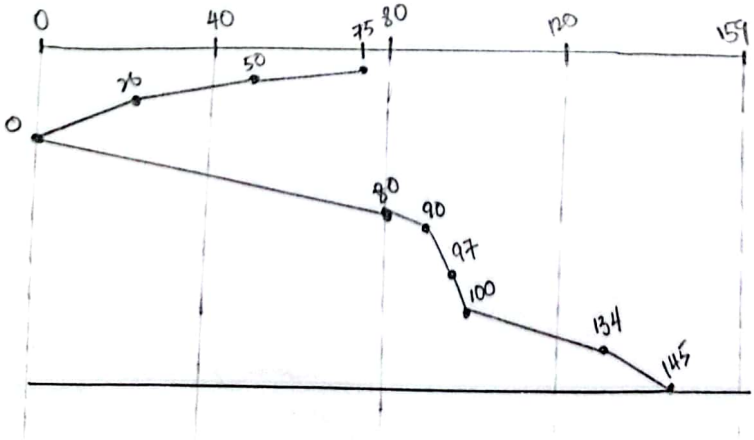
= 293 cylinders - 217 cylinders = 76 cylinders.

The total distance of SCAN is less than C-SCAN about 76 cylinders.

SCAN is faster than C-SCAN.

The disk arm starts at one end of the disk, and moves toward the other end, servicing requests until it gets to the other end of the disk, where the head movement is reserved and servicing continues.

d)



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Question 3

a) i) root / programs / p / F6

root / bin / recorder / recorder / first / F19

ii) mail / F7

All / F17

iii) cd /spell / mail / prf

iv) Relative filename

v)

Owner	Group	public
6	5	1
110	101	001
RW-	R-D	--D

Owner allowed only read and write File F16.
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vi) -rwxr---wx 7 research admin 512 Sep 3 08:30 F16

vii) "public.doc" is a filed owned by "research" with group "admin".
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chmod a+r, a-wx *doc.

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Question 4

a)

	4	3	2	1	2	3	4	7	3	2	1	2	4	5	1	2	1	3	2	1
Frame #0	4	4	4	4				7					7	7		7	1	1		
Frame #1		3	3	3				3					4	4		4	4	3		
Frame #2			2	2				2					2	5		5	5	5		
Frame #3				1				1					1	1		2	2	2		

No. of page fault = 10

b)

	4	3	2	1	2	3	4	7	3	2	1	2	4	5	1	2	1	3	2	1
Frame#0	4	4	4	1			1	1	3	3	3		4	4		4	1	1		
Frame#1		3	3	3			4	4	4	2	2		2	5		5	5	3		
Frame#2			2	2			2	7	7	7	1		1	1		2	2	2		

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c) 4 frames have less 4 page faults than 3 frames by using FIFO. (14 - 10 = 4)

More frame number can reduce page fault amount.

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d)

	4	3	2	1	2	3.	4	7	3	2	1	2	4	5	1	2	1	3	2	1
Frame#0	4	4	4	4				7					4	5						
Frame#1		3	3	3				3					3	3						
Frame#2			2	2				2					2	2						
Frame#3				1				1					1	1						

No. of page fault = 7

e) OPT have less 3 page faults than FIFO (10 - 7 = 3)

The system efficiency of OPT is greater than the system efficiency of FIFO since it has less page fault number compered to FIFO.

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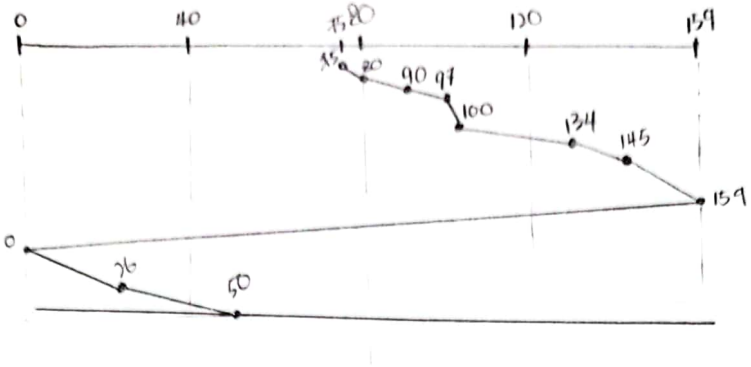
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97, 100, 50, 134, 26, 145, 80, 90

26, 50, 80, 90, 97, 100, 134, 145

a)



b) Total distance

= 159 - 75 + 159 - 0 + 50 - 0

= 84 + 159 + 50

= 293 cylinders.

c)

Total distance

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= 217 cylinders.

Difference

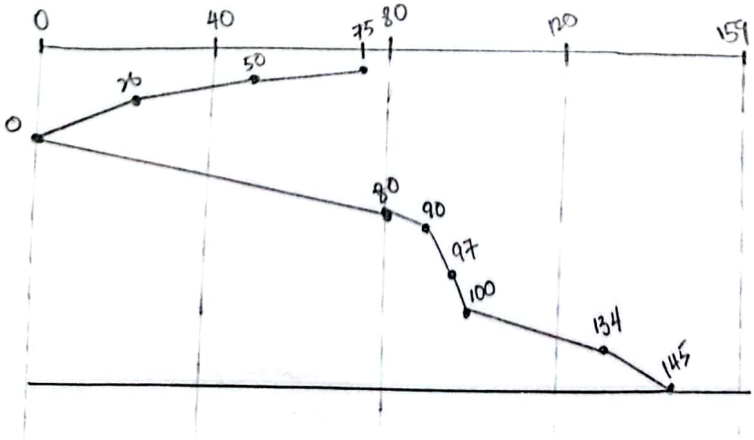
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d)



SECTION B (2013/2014)

STRUCTURED QUESTIONS (50 MARKS)

(**INSTRUCTION**: Please answer all questions in the space provided.)

QUESTION 1 [23 marks]

(a) Given the following Resource-Allocation Graph (RAG) as in Figure 1, identify set for *edges, E*. [3 marks]

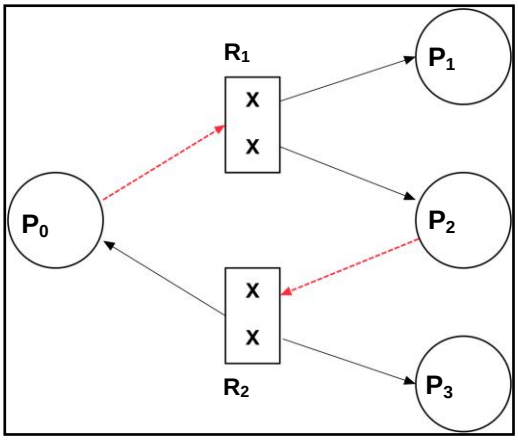


Figure 1: Resource-Allocation Graph

(b) State 2 methods for deadlock recovery. [2 marks]

(c) Consider a system with 4 processes P_0 through P_3, 3 resource types A, B, and C, and the current state at t_0 as shown in Table 1. *Hint: Safety algorithm and Resource-request algorithm in Appendix 1.*

- Resource type A has 11 instances
- Resource type B has 8 instances
- Resource type C has 6 instances

Table 1: Process allocation

Process	Allocation			Max		
	A	**B**	**C**	**A**	**B**	**C**
P_0	2	0	0	4	2	1
P_1	3	1	2	5	3	5
P_2	2	1	1	2	3	2
P_3	1	3	1	1	4	5

i) Calculate the *Need* parameter for each process based on Table 1. [2 marks]

ii) Execute the safety algorithm to determine either the system is in safe or unsafe state and write the safe sequence (if safe state). Show all your workings.

[6 marks]

iii) Assume P_1 request 1 additional resource instance of resource type B and 1 instance of resource type C. Using *resource-request algorithm* and *safety algorithm*, identify if P_1 requests can be fulfilled by determining the new safe sequence. Show all your workings.

[6 marks]

iv) After P_1's request is fulfilled in question (iii), P_3 makes the following request, $Request_3$ = (0, 0, 1). Can the system grant the request of P_3 using resource-request algorithm and safety algorithm? Explain your answer. [4 marks]

QUESTION 2 [20 marks]

(a) In an operating system, dynamic memory partition is implemented.

i) Show how the following set of jobs {Job1: 200 KB, Job2: 280 KB, Job3: 417 KB, and Job4: 500 KB} are allocated inside the main memory. [3 marks]

ii) Referring to Table 2, assume at t_4 Job3 ends and at t_6 Job1 ends producing the following new arrival sequence:

Table 2: Time and job description

Time	Job #	Job Size in KB
t_5	5	200
t_7	6	500

a) Show/draw how these new jobs are allocated into memory space at t_4 till t_7. [4 marks]

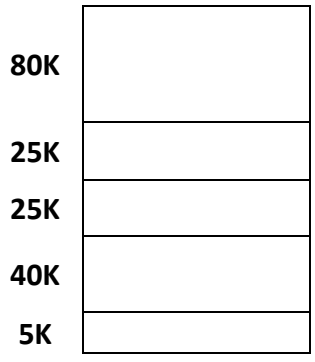
b) Is there any job that cannot be allocated into the memory? If yes, state which job and give a reason. [2 marks]

(b) Given new Jobs in Table 3 and main memory using relocatable dynamic partition in Table 4.

Table 3: Job description

Jobs	Size
1	40K
2	40K
3	20K
4	25K
5	10

Table 4: Memory partition



i) Draw and label each job that will be allocated into the memory and shades all unused partition clearly. [4 marks]

ii) Calculate the total of fragmentation of the memory for the above case. [2 marks]

iii) If another Job 6 with size 35K arrives, can this job be allocated into the memory? Suggest your solution. [5 marks]

QUESTION 3 [8 marks]

(a) Given a paging system with 2^{24} bytes of physical memory; 2^{10} pages of logical address space and page size of 512 bytes.

i) What is the total size of the logical address space. [1 mark]

ii) What is the frame size of the physical memory. [1 mark]

iii) How many frames of the physical memory. [1 mark]

iv) How many number of entry in the page table. [1 mark]

v) Calculate the offset bits for a page of virtual memory. [1 mark]

vi) Draw the frame offset of the physical memory address and page offset of logical memory. [3 marks]

QUESTION 4 [15 marks]

(a) Given the following reference string:

`1 3 2 1 4 3 2 4 1 5 2 4`

For each of the following page replacement algorithms, calculate how many **page faults** will occur if **3 frames** are available:

i) Least Recently Used (LRU) [4 marks]

ii) Optimal Page Replacement (OPT) [4 marks]

iii) Based on your answer in a i) and a ii), which algorithm is better? Justify your answer. [2 marks]

(b) Define what is thrashing**?** [1 mark]

(c) State 2 techniques to prevent thrashing. [2 marks]

(d) Based on Figure 2, explain the upper and lower bound of page fault. [2 marks]

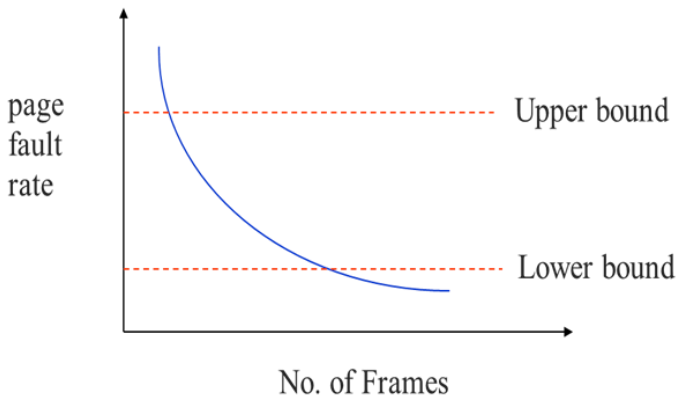


Figure 2: No. of frames vs page faulty rate

QUESTION 5 [24 Marks]

(a) Suppose a user of a UNIX system is currently in the home directory `/home/user1`. If the user issue the command `cd classes/scsr2043` and then the command `pwd`, what will be the user's current directory? Assume all directories exist. [1 mark]

(b) A user issues the Linux command `ls -l` and the following output is displayed on the screen:

```
-rw-r--r-- 1  muhalim  muhalim      0  Dec 30  13:12   fork2.c
drwxr-xr-x 2  muhalim  muhalim   4096  Dec 30  11:52   ResultC
```

i) Is `ResultC` a directory or a file? [1 mark]

ii) What is the access right of the owner of `fork2.c` [1 mark]

ii) Give a Linux command that will enable a group of users that share `fork2.c` to write to the file. [1 mark]

(c) Match the list of terms given below to each of the descriptions: [2 marks]

Access Control List	Access Control Matrix	Two-level directory
Contiguous Allocation	Indexed Allocation	Bit Map / Bit Vector
C-SCAN	SSTF	RAID	Data Stripping

i) A technique for spreading data over multiple disks.

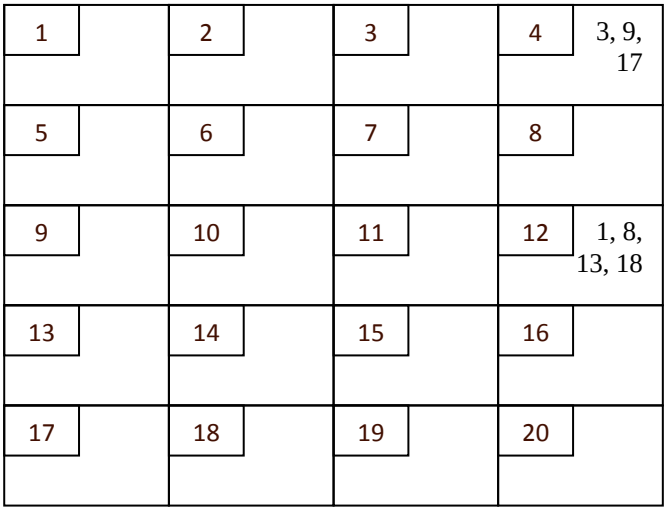
ii) A file protection mechanism that uses much more storage space as files and users increase.

iii) Provides easy direct access to records in a file (Every parts of a file are stored in same compact area).

iv) A technique to manage free spaces in disk.

(d) List 3 factors which determine how long it takes to read or write a disk block. Provide the formula to compute Access time. [2 marks]

(e) Figure 3 shows a diagram of disk storage containing two files, `File1` and `File2`:



File	Index block
File1	12
File2	4

Figure 3: Disk storage

i. Add arrows in the diagram if `File1` and `File2` start at block 12 and block 4 respectively. [3 marks]

ii. State the name of disk allocation technique that is being used. [1 mark]

(f) Figure 4 shows a diagram of disk storage containing two files, `File1` and `File2`:

Directory

File	Address	Size	Next
	1	512	
File1 (1)	2	512	8
File2 (1)	3	512	6
	. .	. .	
File2 (2)	6	512	11
	7	512	
File1 (2)	8	512	20
	. .	. .	
File2 (3)	11	512	-
	. .	. .	
File1 (3)	20	512	-

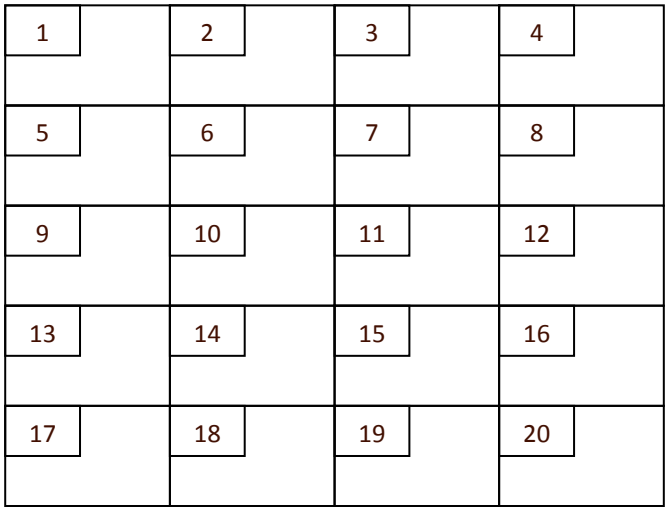


Figure 4: Disk Storage

i) Add arrows in the diagram according to the directory information for `File1` and `File2`. [3 marks]

ii) State the name of disk allocation technique that is being used. [1 mark]

(g) Suppose disk storage has block size of 4KB and disk size of 2 GB. Calculate total space needed for the bit vector. [2 marks]

(h) Assume the head of a disk is receiving track requests waiting in the queue in the following order:

98, 183, 65, 14, 122, 100, 20, 35

If the head starts at track 70 and going towards inner direction, show the movement path of the read-write head arm if the following seek strategies used:

i) C-SCAN [3 marks]

0 14 20 35 65 98 100 122 183 199

---

ii) C-LOOK [3 marks]

0 14 20 35 65 98 100 122 183 199

------- End of Question -------

GOOD LUCK !!

APPENDIX 1

Safety Algorithm

1. Let ***Work*** and ***Finish*** be vectors of length ***m*** and ***n***, respectively.
   Initialize:
   Work = Available
   Finish[i] = false for ***i***= 1, 2, …, ***n***
2. Find an index ***i*** such that both
   Finish[i] == false
   $Need_i \leq Work$
   If no such ***i*** exists, go step 4.
3. $Work = Work + Allocation_i$
   Finish[i] == true
   Go to step 2.
4. If ***Finish[i] == true*** for all ***i***→ then the system in safe state.

Resource-Request Algorithm

1. If $Request_i \leq Need_i$, go to step 2.
   Otherwise, raise an error condition since the process has exceeded its maximum claim.

2. If $Request_i \leq Available$, go to step 3.
   Otherwise, P_i must wait since the resources are not available.

3. Have the system pretend to have allocated the requested resources to process P_i by modifying the state as follows:

   $Available := Available - Request_i$
   $Allocation_i := Allocation_i + Request_i$
   $Need_i := Need_i - Request_i$

4. Execute the **safety algorithm** (Figure B.5) on a **new state**. If the resulting state is safe, allocate the resources, else delay the request.

SECTION B (2014/2015)

STRUCTURED QUESTIONS (80 MARKS)

(**INSTRUCTION**: Please answer all questions in the space provided.)

QUESTION 1 (20 MARKS)

a. You are given the following set of edges. Use it to answer the following questions.

Edge, E = {P0 → R1, P1 → R1, R1→P2, P2→R2, R2→P1, R3→P0}

i. Draw the Resource Allocation Graph. [3]

ii. Using the RAG in (i) for the scenarios given below, please explain the possibility of deadlock happening (if deadlock happens). [4]

Scenario A

R1 has 2 instances, R2 has 2 instances, R3 has 1 instance

Scenario B

R1 has 1 instance, R2 has 2 instances, R3 has 2 instances

b. Consider the following system has three processes, P0, P1 and P2, and with three resource types, A, B and C. At time t_0, the resource allocation state of the system is as shown below:

	Max			**Allocation**		
	A	**B**	**C**	**A**	**B**	**C**
P0	4	2	3	2	1	1
P1	3	1	6	1	0	2
P2	6	3	3	3	2	2

Available		
A	**B**	**C**
2	1	2

i. Calculate the total number of resource for type A, B and C. Show your calculation. [2]

ii. What is the content of the Need matrix ? [2]

iii. Determine whether the system is in **safe** or **unsafe state** at t_0. [5]

iv. If at t_1, P2 request for (1,1,1), should the system satisfy the request of P2? Show your works. [4]

QUESTION 2 (22 MARKS)

a. In an operating system, assume memory is allocated as specified in Figure 1.

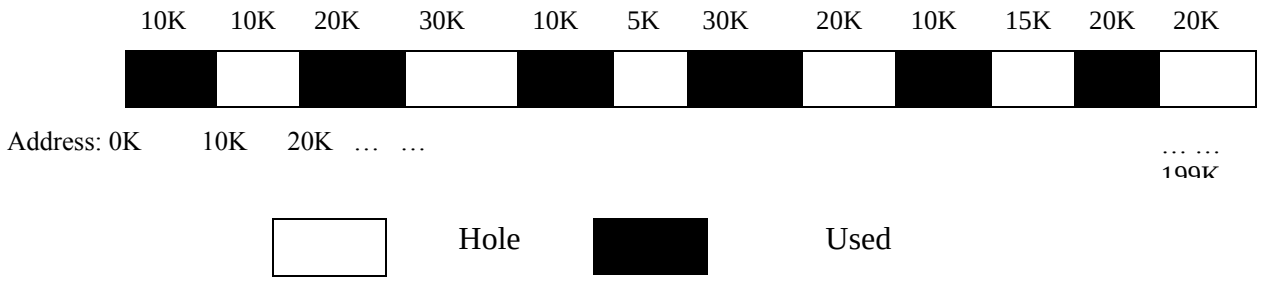


Figure 1: Variable partition memory allocation.

i. Show how the following set of jobs {J1 - 20KB, J2 - 10KB, J3 - 5KB} (in that order) are allocated inside the main memory using **first-fit allocation**. At what starting address will each job be allocated? [3]

ii. Show how the following set of jobs {J1 - 20KB, J2 - 10KB, J3 - 5KB} (in that order) are allocated inside the main memory using **best-fit allocation**. At what starting address will each job be allocated? [3]

b. Given new jobs Job 6 until Job 10 in Figure 2 and main memory using *relocatable dynamic partition* in Figure 3. Consider Job 1 until 5 end at t_{10}.

Jobs	Size
6	40K
7	40K
8	20K
9	25K
10	10K

Figure 2: Job description

80K	Job 1
25K	Job 2
25K	Job 3
40K	Job 4
5K	Job 5

Figure 3: Memory partition

i. Redraw and label each job that will be allocated into the memory after t_{10}. Shades all unused partition clearly. [4]

ii. Calculate the total fragmentation of the memory for case (i). [2]

iii. If another Job 11 with size 35K arrives, can this job be allocated into the memory? Suggest your solution. [1]

c. What are the **TWO** major differences between paging and segmentation? [2]

d. Given a paging system with 2^{24} bytes of physical memory; 2^{10} pages of logical address space and page size of 512 bytes.

i. What is the total size of the logical address space. [1]
ii. What is the frame size of the physical memory. [1]
iii. How many frames of the physical memory. [1]
iv. How many number of entry in the page table. [1]
v. Calculate the offset bits for a page of virtual memory. [1]
vi. Draw the frame offset of the physical memory address and page offset of logical memory. [2]

QUESTION 3 (15 MARKS)

a. A page replacement algorithm should minimize the number of page faults. Consider the following page reference string:

1, 2, 3, 4, 5, 3, 4, 1, 6, 7, 8, 7, 8, 9, 7, 8, 9, 5, 4, 5, 4, 2

Assuming demand paging with **FOUR** page frames, how many page faults would occur for the following replacement algorithms? Show all your works.

i. FIFO [3]

ii. OPT [4]

iii. Compare and discuss both algorithms in term of page faults occurred? Which do you think is better? [2]

b. In paging, thrashing may occur.

i. What is the cause of thrashing? [2]

ii. How does the system detect thrashing? [2]

iii. Once it detects thrashing, what can the system do to eliminate this problem? [2]

QUESTION 4 (23 MARKS)

a. Indicate whether statements below are **TRUE** or **FALSE** statements, according to the Unix files access control as shown in Figure 4.

```
-r-xr--r-x  Ali   Ali        2.6K  Feb 16 2011      Untitled
-rw-r-xr--  Abu   Student    436B  Aug 28 2011      README
drwxr-xr-x  Ali   Student    197B  Aug 28 2011      Notes
```

Figure 4: Unix file access control

i. Only Untitled and README are files. [1]
ii. Notes is a folder that is shared among users of the system. [1]
iii. Owner of the README file is Student. [1]
iv. Untitled file is editable by Ali. [1]
v. README is only editable by Abu. [1]

b. Assuming Figure 5 shows a storage map of a disk with capacity of 400KB. The disk has 20 storage blocks and size of a storage block is 20KB. Two files will be stored in the storage disk that are **File1** and **File2**. Size of File1 is **58KB**, while the size of File2 is **36KB**. Assuming the operating system is using **Indexed Storage Allocation**. Each storage block can only store 3 storage block addresses.

i. Fill in the Index Block Number column in Directory File, and update the Storage Map accordingly for File1. Consider Index block for File 1 is **block 3**. [2]
ii. Fill in the Index Block Number column in Directory File, and update the Storage Map accordingly for File2. Consider Index block for File 2 is **block 18**. [2]

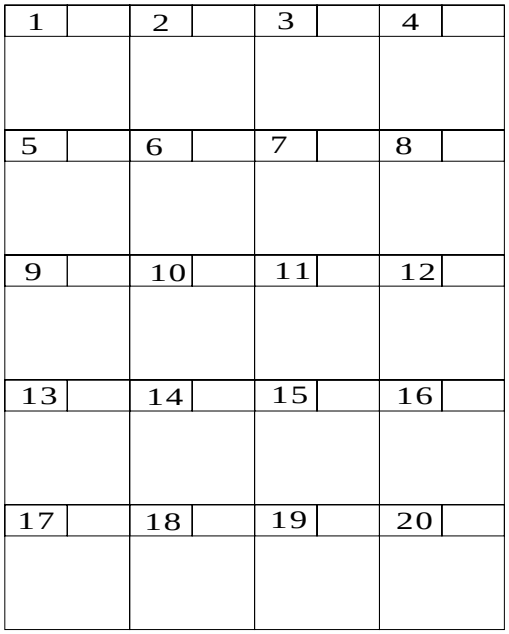


Figure 5: Storage Map

Filename	File Size (KB)	Index Block Number
File1	58	
File2	36	

Directory File

c. Figure 6 shows a bit map for free space management for a hypothetical Operating System whereby bit **0** indicates that the block is used and bit **1** indicates that the block is **free**.

Block No	**0**	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**13**	**14**	**15**
Bit Map	0	1	1	0	1	0	0	1	0	1	1	1	0	1	0	0

Figure 6: Storage Map

Suppose that the size of a block is 15KB.

i. List the block numbers that are allocated to a file named DATA.txt. The size of the DATA.txt is 64KB. [3]

ii. Is there any block that suffers from fragmentation in storing the DATA.txt? If your answer is a YES, state the block number that contains the fragmentation. [2]

d. On a disk with 900 tracks: numbered 0 to 899. The disk queue in FIFO order contains requests on the following tracks:

142, 778, 292, 575, 205, 376

Starting from the current head location **545**, sketch figure to illustrate the head movement (track traveled) to serve all the requests based on

i. **SSTF** seek strategy. [3]

ii. **C-LOOK** seek strategy. The head movement direction is outer to inner. [3]

iii. According to the above scenario (i) and (ii), which seek strategy is better between SSTF and C-LOOK? Briefly justify your answer. [3]

SECTION B (2015/2016)

STRUCTURED QUESTIONS (80 MARKS)

(**INSTRUCTION**: Please answer all questions in the answer booklet)

QUESTION 1 [25 MARKS]

a). Answer the following questions based on the basic concepts of memory management for operating systems.

i. The management of main memory is critical, thus the entire system performance is dependent on two (2) items/factors. Give the **two (2)** items/factors. [2]

ii. Describe the fundamental difference between **logical address** and **physical address**? [2]

iii. A process can be swapped temporarily out of memory to a backing store and then brought back into memory for continued execution. In the simplest term, what does **backing store** here refers to? [2]

iv. When **a process is being swap out** and another desired **process is being swap in by the dispatcher** for execution, what does it actually means? [2]

v. Swapping to free up memory is constrained by some factors. **Can a process waiting for I/O operation be swapped?** Furthermore, what is the **primary selection criterion to swap out** a process? [2]

b). The following questions deals specifically with the **fixed partitions memory scheme**.

i. As shown in figure below, there are **nine (9) fixed partitions of different sizes,** and there are several jobs waiting to be loaded into memory for execution. How would the **First-fit**, **Best-fit** and **Worst-fit** algorithms place all the jobs into the memory partitions? [8]

Memory Partitions

PARTITION NO.	PARTITION SIZE(KB)
1	50
2	110
3	80
4	150
5	100
6	90
7	130
8	160
9	60

Jobs List

JOBS	SIZE OF JOB (KB)
A	60
B	120
C	40
D	30
E	80
Z	100

Figure: Jobs Allocation into Fixed Partitions

Allocation Strategy / Partition	1	2	3	4	5	6	7	8	9
Next-fit (This has been done for you)		**A**		**B**	**C**	**D**	**E**	**Z**	
First-fit									
Best-fit									
Worst-fit									

ii. Calculate the **size of internal fragmentations incurred for each job** and then compute the **total size of fragmentation for First-fit and Best-fit** job allocation schemes. [4]

JOBS	First-fit internal fragmentation	Best-fit internal fragmentation
A		
B		
C		
D		
E		
Z		
Total size of internal fragmentation:		

iii. For all the jobs allocation schemes above, **is there any job that needs to wait from entering a partition?** State the job and allocation scheme. [1]

iv. **Rank all the three (3) jobs allocation** schemes above (First-fit, Best-fit, Worst-fit) in terms of how efficiently they use memory. [2]

Rank 1:

Rank 2:

Rank 3:

QUESTION 2 [10 MARKS]

a). State **TRUE** or **FALSE** for the following statements. [3]

Statements	TRUE/ =FALSE
Paged memory allocation divides each incoming job into pages of equal size.	
The disadvantage of paged memory is that it allows job allocation in non contiguous memory.	
Page table consist of page number and corresponding page frame memory address.	

b). Given a paging system with 2^{32} bytes of physical memory; **2^{10} pages** of logical address space and **page size of 4096 bytes**. [7]

i. What is the total size of the logical address space.

ii. What is the frame size of the physical memory.

iii. How many frames of the physical memory.

iv. How many number of entry in the page table.

v. Calculate the offset bits for a page of virtual memory.

vi. Draw the frame offset of the physical memory address and page offset of logical memory.

QUESTION 3 [15 MARKS]

CPU is executing a loop for an array implementation and a series of memory locations are accessed repeatedly. Given the following **reference string**:

1 2 3 4 1 2 3 4 1 2

a). Apply **LRU page replacement** algorithm and **calculate how many page faults** will occur. Assume **three (3) page frames** are used. **Show all your workings.** [4]

b). Apply **OPT page replacement** algorithm and **calculate how many page faults** will occur. Assume **three (3) page frames** are used. **Show all your workings.** [4]

c). Which algorithm has the **best performance in term of page faults occurred**? **Justify your answer.** [1]

d). **State 2 methods** used in LRU implementation. [2]

e). Does increasing the **page frames to four (4) cause Belady's anomaly** effect on LRU? **Justify your answer.** [4]

QUESTION 4 [10 MARKS]

a). Arrange READ logical command statements in correct sequence when interacting with file manager. [2]

Statements	Sequence
Wait for rotational delay (sector containing record passes under read/write head)	
Send flag indicating free device for another request	
Activate appropriate read/write head and read record	
Transfer record to main memory	
Move read/write heads to record cylinder	**1**

b). Indicate whether statements below are **TRUE** or **FALSE** according to the Unix files access control list as shown in figure below. [6]

```
-rwxr--r-x   Ali    Ali          2.6K      Feb  16  2011      Untitled
-rw-r-xr-x   Abu   Student     436B   Aug  28  2011    README
drwxrwxr-x   Ali   Student     197B   Aug  28  2011    Notes
```

Figure: Unix file access control

i. Untitled and README are directory.

ii. README group access right is read only.

iii. Notes is a folder that is shared among Student group.

iv. Owner of the README file is Student.

v. Untitled file can be edited by Ali.

vi. README can be executed by public.

c). The following figure below shows some output from file-system related commands on a local Linux system. Use this output and your knowledge of Linux file systems to answer the following questions. [2]

```
[carey@csl]$ df
```

Filesystem	1K-blocks	Used	Available	Use%	Mounted on
/dev/sda2	20315844	7513824	11753380	39%	/
/dev/sda5	10581704	5181092	4854404	52%	/tmp
/dev/sda1	1019208	50356	916244	6%	/boot
tmpfs	1684784	77164	1607620	5%	/dev/shm
nsi:/export/research	247709760	203879488	31247360	87%	/home/research
nse:/export/scratch	3361364788	528637368	2661979928	17%	/home/scratch
nsj:/export/proj/dsl	485097928	184513976	275942416	41%	/home/dsl
nsg:/export/ug	381885660	79136396	283350608	22%	/home/ugc
nsf:/export/ug	81885660	88768580	273718424	25%	/home/ugb
nsh:/export/grads	789574392	507946744	241519616	68%	/home/grads
nsb:/export/ug	381885660	76958700	285528304	22%	/home/uga

Figure X: Unix file-system output

i. How many different file systems are accessible on this Linux system?

ii. Which file system is the fullest (in terms of percent occupancy)?

iii. Which file system has the largest physical storage capacity?

iv. Which file system has the fewest bytes currently stored?

QUESTION 5 [10 MARKS]

Figure below shows a storage map of a disk. The disk has **twenty (20) storage blocks and size of a storage block is 256KB**. Two files will be stored in the storage disk are **DATA1** and **DATA2**. Size of DATA1 is **718KB**, while the size of DATA2 is **300KB**.

Assuming the operating system is using **Indexed Storage Allocation**, each storage block can only **store 4 block addresses**.

1		2		3		4	
5		6		7		8	
9		10		11		12	
13		14		15		16	
17		18		19		20	

Figure Storage Map

Filename	File Size (KB)	Index Block Number
DATA1	718	
DATA2	300	

Directory File

a). What is the **maximum capacity of the disk**? [2]

b). Fill in the Index Block Number column in Directory File and update the Storage Map accordingly for DATA1. Consider index block for DATA1 is **block 6** and use blocks with the **lowest number value** for DATA1 storage map. [3]

c). Fill in the Index Block Number column in Directory File and update the Storage Map accordingly for `DATA2`. Consider Index block for `DATA2` is **block 16** and use blocks with the **highest number value** for `DATA2` storage map. [3]

d). **Is there any block** that suffers **from fragmentation in storing the `DATA2` file**? If your answer is a YES, **state the block number** and **calculate the size of fragmentation** that incurred in the block. [2]

QUESTION 6 [10 MARKS]

On a disk with 800 tracks: **numbered from 0 to 799**. The disk queue in FIFO contains the following track requests:

233, 75, 744, 300, 680, 190

Starting from the current head location **600**, sketch figure to illustrate the head movement (track traveled) to serve all the requests based on:

a). **SSTF** seek strategy. [4]

0 75 190 233 300 600 680 744 799

b). **C-SCAN** seek strategy. The head movement direction is **outer to inner**. [4]

0 75 190 233 300 600 680 744 799

c). According to the above scenario in **(a) and (b)**, which seek **strategy is better** between **SSTF** and **C-SCAN**? **Briefly justify your answer**. [2]